Model Evaluation & Knowledge Tracing

Machine Learning for Behavioral Data

March 10, 2026

EPFL

IC

Today's Topic

Week	Lecture/Lab
1	Data Exploration
2	Regression
3	Classification + Model Evaluation
4	**Evaluation + Knowledge Tracing**
5	Time Series Prediction
6	Time Series Prediction
7	Unsupervised Learning
8	Spring Break

Getting ready for today's lecture…

- Join us on SpeakUp

https://go.epfl.ch/mlbd-speakup

Short quiz about the past…

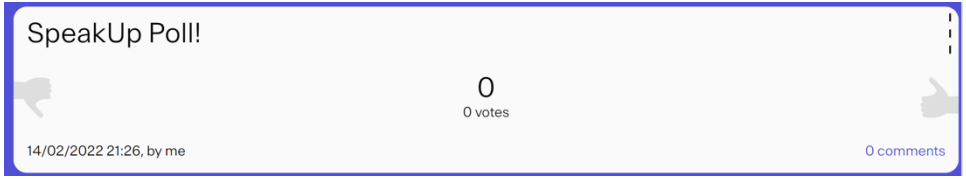
We are going to predict the species of an Iris Flower. The dependent variable (species) contains three possible values: Setosa, Versicolor, and Virginica. Which classification methods are appropriate for this task?

a) Decision Tree

b) k-Nearest Neighbor

c) Logistic Regression

d) Random Forest



https://go.epfl.ch/mlbd-speakup

Short quiz about the past…

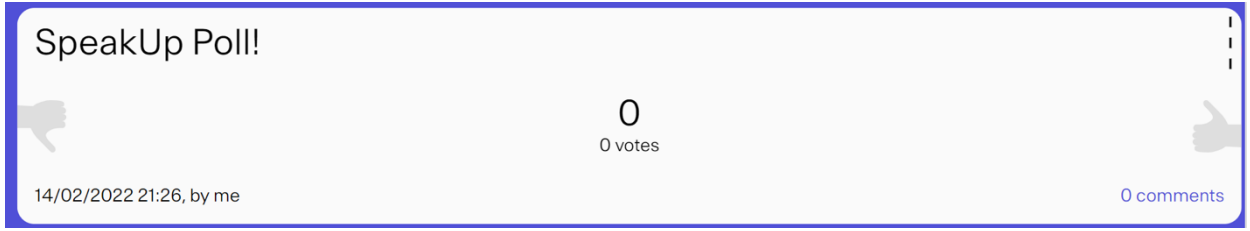
Which of the following statements about k-fold cross validation are wrong? N denotes the number of samples in the data set, k the number of folds:

a) k must always be smaller than N.

b) The smaller k is, the more expensive it is to compute the error.

c) Cross validation can be used to tune model hyperparameters.

d) Cross validation is not a valid method for computing the generalization error of a model.



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Agenda

- **Model Evaluation:**
  - **Model Selection and Assessment**
  - Error Metrics and Reporting Results
- Time Series – Bayesian Knowledge Tracing

Assessment & Selection

1. Train, Validate, Test
2. **Resampling Methods**
   - Cross Validation
   - **Bootstrapping**
3. Information Criteria

Bootstrapping

- General tool for assessing statistical accuracy
- Basic idea:
  - Given: training data set $\boldsymbol{Z} = \{z_1, \dots, z_N\}$ with $z_i = (\boldsymbol{x_i}, y_i)$, where N is the number of samples
  - Randomly draw N pairs $(\boldsymbol{x_i}, y_i)$ with replacement from the training data set
  - Repeat B times (e.g., $B = 100$) -> B bootstrap data sets
  - Fit model to each bootstrap data set, observe behavior across bootstrap data sets

Bootstrapping

- We are interested in $S(Z)$
- $S(Z)$ can be any quantity computed from the data Z, e.g., the prediction at some input point
- We want to compute an aspect of the distribution of $S(Z)$ (e.g., the variance)

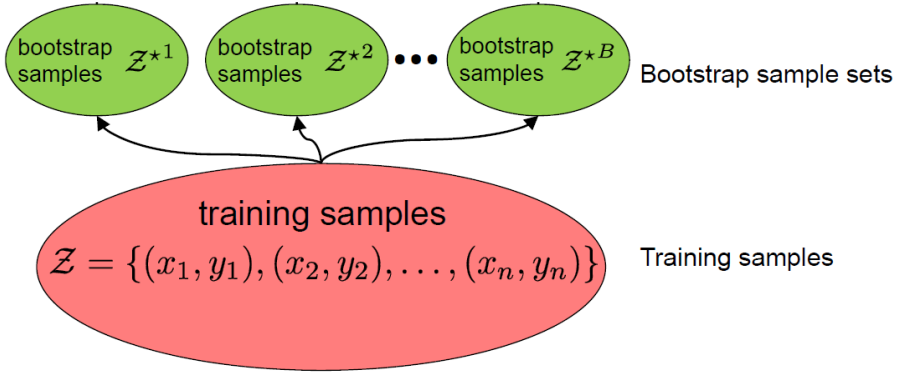
training samples

$$\mathcal{Z} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$$

Training samples

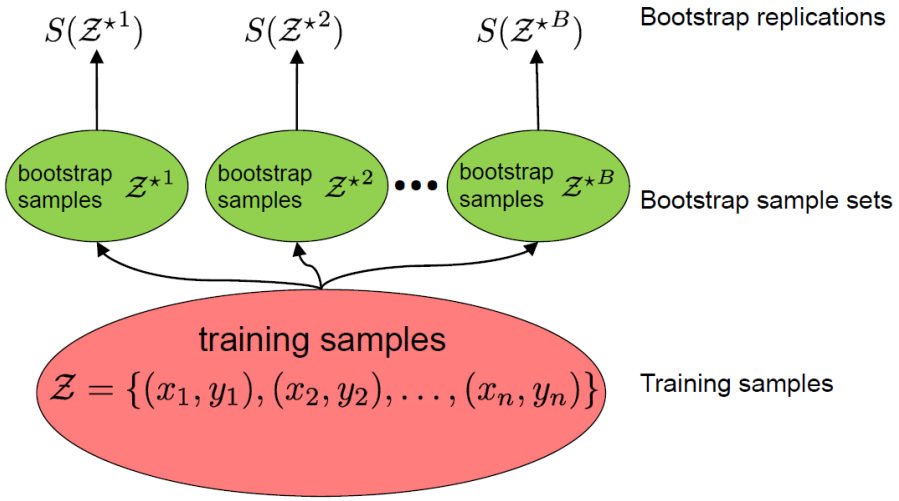
[Illustration credit: ML lecture, ETH Zurich]

Bootstrapping



[Illustration credit: ML lecture, ETH Zurich]

Bootstrapping



[Illustration credit: ML lecture, ETH Zurich]

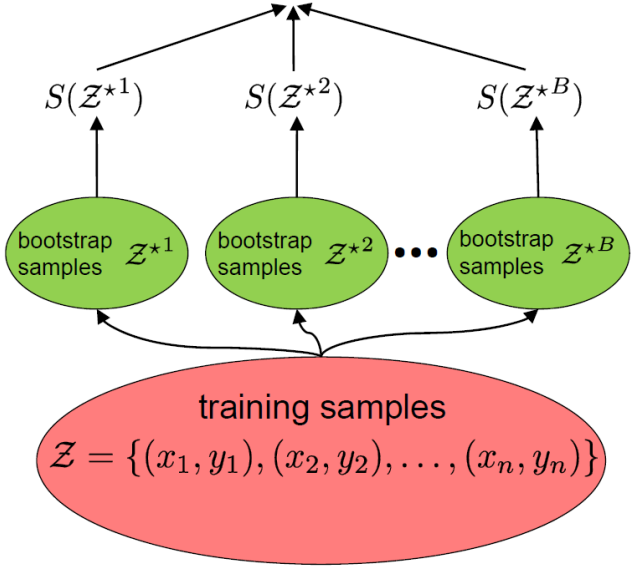
Bootstrapping

$$\overline{S} = \frac{1}{B}\sum_{b\leq B} S(\mathcal{Z}^{\star b})$$

Replication mean

$$\sigma^2(S) = \frac{1}{B-1}\sum_{b\leq B}\left(S(\mathcal{Z}^{\star b}) - \overline{S}\right)^2$$

Replication variance



Bootstrap replications

Bootstrap sample sets

Training samples

[Illustration credit: ML lecture, ETH Zurich]

Bootstrapping for prediction

- Idea: leave-one-out bootstrap

$$\widehat{Err}^{(1)} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{|C^{-i}|}\sum_{b\in C^{-i}} L(y_i, f^{*b}(\boldsymbol{x_i}))$$

where C^{-i} denotes the set of indices of bootstrap samples b, that do **not** contain observation i

.632 bootstrap error

- It can be shown that the average number of distinct samples in each bootstrap sample b is about $0.632 \cdot N$
- $\widehat{Err}^{(1)}$ will therefore tend to overestimate the true loss
- Solution:

$$\widehat{Err}^{(.632)} = .368 \cdot \overline{err} + .632 \cdot \widehat{Err}^{(1)}$$

Assessment & Selection

1. Train, Validate, Test
2. Resampling Methods
3. **Information Criteria**
   - AIC
   - BIC

Information Criteria

- The training error is an overly optimistic estimate of the *expected loss*
- Information criteria try to estimate the *optimism* in the training error and correct for it

Akaike Information Criterion (AIC)

$$\boxed{AIC = -\frac{2}{N} \cdot loglik + 2 \cdot \frac{d}{N}}$$

- d is the number of parameters of our model f
- $loglik$ is the log-likelihood (logarithm of the likelihood) of the sample data T given our model f
- N is the number of samples in the data set, i.e. $|T| = N$

Bayesian Information Criterion (BIC)

$$\boxed{BIC = -2 \cdot loglik + \log(N) \cdot d}$$

- d is the number of parameters of our model f
- $loglik$ is the log-likelihood (logarithm of the likelihood) of the sample data T given our model f
- N is the number of samples in the data set, i.e. $|T| = N$
- BIC penalizes complex models more heavily than AIC

Information Criteria - Considerations

- Applicable for models where the fitting is done under a maximum likelihood setting
- *Effective* number of parameters:
  - Choosing the best fitting model with d features, can result into more than d parameters being fit (e.g., regularization)
  - Determining the *effective* number of parameters can be difficult for more complex models (e.g., trees)

What method should we choose?

Depends on the purpose of the model:

- Interpretation: information criteria are useful for unsupervised cases or when the model is built for interpretation purposes (e.g., which features of the regression model explain the data better?)
- Prediction:
  - All presented methods (train-test, cross validation, bootstrap) are reasonable choices
  - We can combine them freely to do model selection and assessment
  - In practice: while bootstrap provides accurate estimates of test error, it also requires a high amount of extra work

Agenda

- **Model Evaluation:**
  - Model Selection and Assessment
  - **Error Metrics and Reporting Results**
- Time Series – Bayesian Knowledge Tracing

Regression: R^2

$$\boxed{R^2 = 1 - \frac{SS_{res}}{SS_{tot}}}$$

where $SS_{res} = \sum_i (y_i - f(x_i))^2$ and $SS_{tot} = \sum_i (y_i - \bar{y})^2$

- Can be interpreted as the fraction of explained variability of the data
- Often used when the goal is *interpretation*
- Often used in the fields of Psychometrics, Learning Sciences, Psychology, etc.

Regression: MAE and RMSE

- Mean absolute error:

$$\boxed{MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - f(x_i)|}$$

- Root mean squared error:

$$\boxed{RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - f(x_i))^2}}$$

- Often used when the goal is *prediction*
- **RMSE is largely preferred to MAE**

Hypothetical Example

- Given:
  - Student giving correct answers 70% of the time
  - Model A: predicts correct 70% of the time
  - Model B: predicts 100% correctness

MAE: Model B is better

- 70% of the time the student gives a correct answer (response = 1)
  - Model A: absolute error = 0.3
  - Model B: absolute error = 0.0
- 30% of the time the student answers wrong (response = 0)
  - Model A: absolute error = 0.7
  - Model B: absolute error = 1.0
- $MAE_A = 0.42,\ MAE_B = 0.30$

RMSE: Model A is better

- $RMSE_A \approx 0.46$
- $RMSE_B \approx 0{,}55$

- $\boldsymbol{RMSE}$ **penalizes large errors heavier**

Classification: Accuracy

$$Acc = \frac{|agreements|}{N}$$

where $agreements$ means that the predicted outcome is equal to the observed outcome

- Easy to interpret
- Works for binary and multi-class
- General agreement across fields: accuracy is not a good standalone performance metric

Why?

Classification: Balanced accuracy

$$\boxed{Acc_{bal} = \frac{1}{|C|} \cdot \sum_{c \in C} Acc_c}$$

where $|C|$ denotes the number of classes

- Easy to interpret: average accuracy over all classes
- Takes into account class imbalance
- Works also for the multi-class case

Classification: Confusion Matrix

		True Outcome	
		Positive	Negative
Predicted Outcome	Positive	True Positive (tp)	False Positive (fp)
	Negative	False Negative (fn)	True Negative (tn)

Confusion matrix for a binary classification task (two classes denoted as positive and negative class)

Classification: Specificity, sensitivity,…

		True Outcome	
		Positive	Negative
Predicted Outcome	Positive	True Positive (tp)	False Positive (fp)
	Negative	False Negative (fn)	True Negative (tn)

$$specificity = \frac{tn}{tn + fp} \qquad sensitivity = \frac{tp}{tp + fn}$$

Classification: Specificity, sensitivity,…

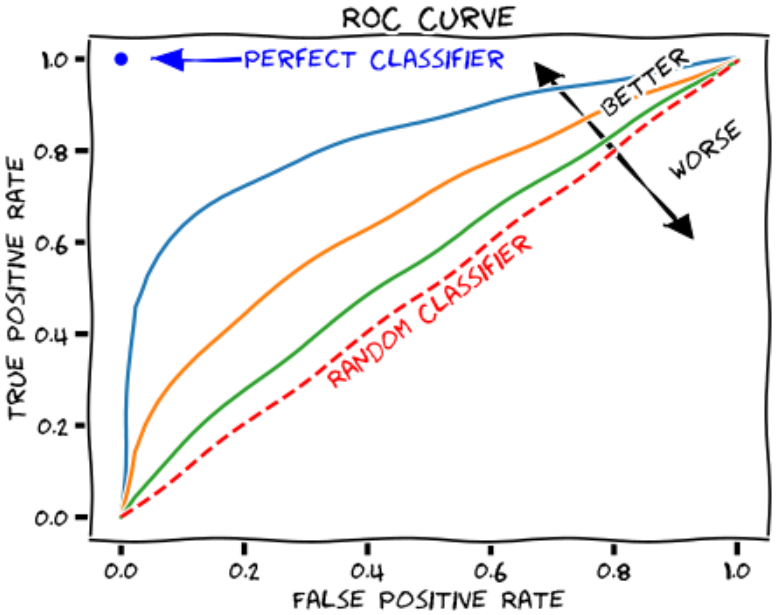
		True Outcome	
		Positive	Negative
Predicted Outcome	Positive	True Positive (tp)	False Positive (fp)
	Negative	False Negative (fn)	True Negative (tn)

$$specificity = \frac{tn}{tn + fp} \qquad sensitivity = \frac{tp}{tp + fn}$$

- These metrics are used in addition to accuracy
- Sensitivity and specificity are often used in the fields of Psychology, Learning Sciences, etc.

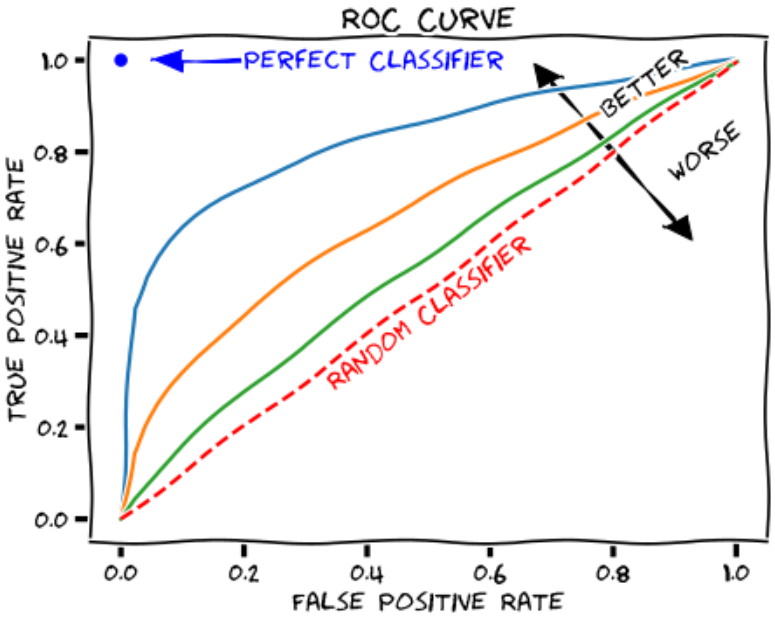
ROC curve

$$TPR = \frac{tp}{P}$$



$$FPR = \frac{fp}{N}$$

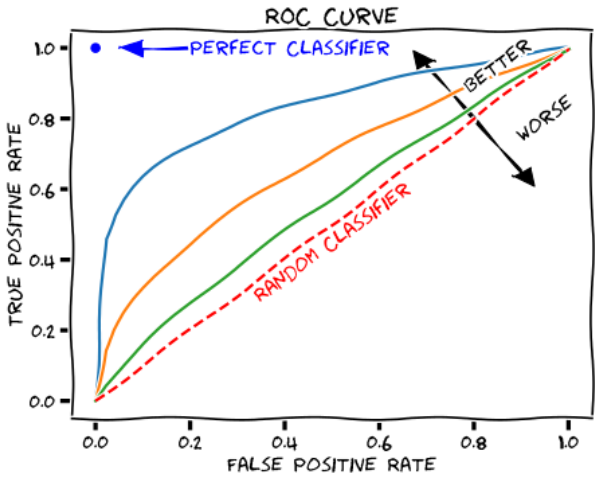
Computing the ROC curve



- Given a classifier $f(\boldsymbol{x})$ predicting some type of score (e.g., a probability)
- Choose a threshold t:
  - $f(x_i) \geq t$: predict positive class
  - $f(x_i) < t$: predict negative class
- Compute TPR and FPR given t
- Repeat for different thresholds (e.g., in case of probabilities choose $t = 0, 0.1, \ldots 1$)

Area under the ROC curve (AUC)

- The AUC denotes the area under the ROC curve
- A perfect classifier has an AUC of 1
- A random classifier has an AUC of 0.5
- Often used as a performance metric in more technical fields (e.g., educational data mining)
- The AUC can be extended to the multi-class case by considering all possible pairs of classes



Classification: Do's and Don'ts

- Do:
  - Carefully think about the choice of metric (or combination)
  - Some ideas:
    - Use accuracy plus sensitivity and specificity [binary]
    - Use (balanced) accuracy plus AUC [binary + multi-class]
    - Use just AUC [binary + multi-class]
- Don't:
  - Use accuracy as a standalone metric
  - Compute "all" possible metrics that come to your mind

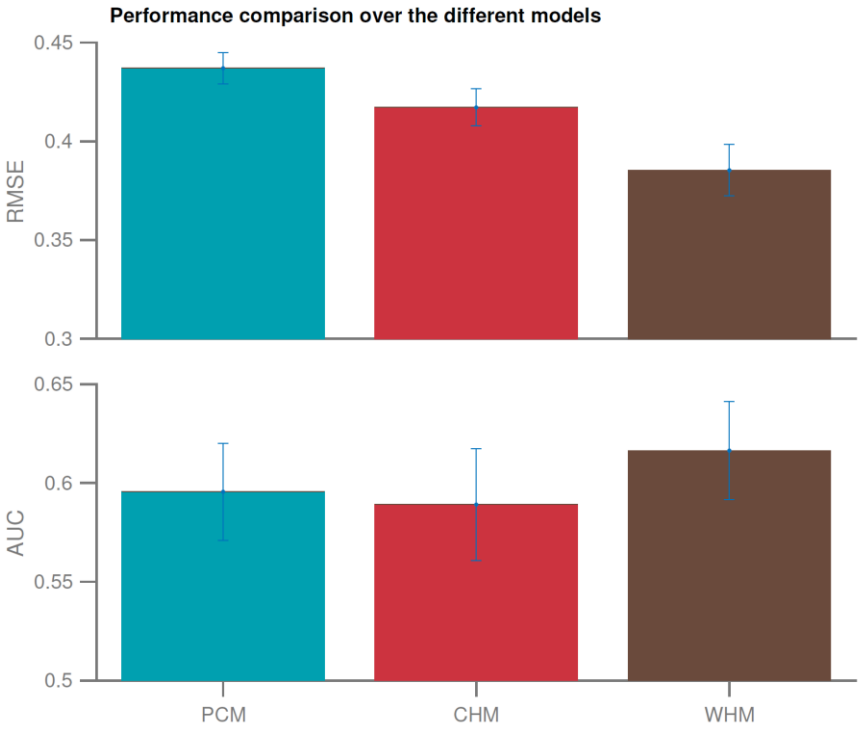
Back to the question: is my model any good?

- A friend tells you that he has found a great model for early prediction of drop out in online courses
- She gives you the following information:
  - The data set stems from N students taking course c
  - 30% of the students dropped the course
  - She has evaluated her model using cross validation
  - The accuracy of the model is 0.8 and the AUC is 0.83

Provide comparisons!

- Reporting performance metrics for a model f on a data set T is not enough: achievable performance (what is good?) depends on the data set!
- We need to provide comparisons to **baseline** models:
  - **Minimum** baseline: compare to a random model
  - Additional comparisons, depending on the goal:
    - Suggestion of a new type of model structure for a problem: need to compare to previously suggested structures
    - Suggestion of new features for a problem: need to compare performance between a model using and not using the new features

Quantification of Uncertainty



- We usually report the *mean* prediction error over all samples
- We should quantify the uncertainty of the performance metric!
- Uncertainty can be computed across samples where applicable (or across folds)
- Error bars can denote
  - Standard deviation
  - Standard error ($\sqrt{\sigma^2/N}$)

[Figure from Käser et al., LAK 2017]

Summary: Pipeline

Design/choose an appropriate learning algorithm and features

- Problem (e.g., classification or regression)
- Data set (e.g., size, type)
- Task (e.g., dropout prediction)

Summary: Pipeline

Design/choose an appropriate learning algorithm and features

Select evaluation method

- Data set (e.g., size and type)
- Learning algorithm (e.g., model complexity)
- Goal (e.g., interpretation or prediction)
- Potential application (e.g., generalization to new users -> stratification)

Summary: Pipeline

Design/choose an appropriate learning algorithm and features

↓

Select evaluation method

↓

Choose appropriate performance metrics

- Problem (e.g., classification or regression)
- Learning algorithm (e.g., model type)
- Potential application (i.e., what is important?)

Summary: Pipeline

Design/choose an appropriate learning algorithm and features

↓

Select evaluation method

↓

Choose appropriate performance metrics

↓

Select baseline approaches for comparison

What do I want to show/prove, i.e. what is my claim (e.g., suggestion of new neural network architecture, suggestion of new features for a specific problem)?

Summary: Pipeline

Design/choose an appropriate learning algorithm and features

↓

Select evaluation method

↓

Choose appropriate performance metrics

↓

Select baseline approaches for comparison

↓

Report your results providing error bars

Summary: Pipeline

Design/choose an appropriate learning algorithm and features

↓

Select evaluation method

↓

Choose appropriate performance metrics

↓

Select baseline approaches for comparison

↓

Report your results providing error bars

There are many ways to solve a given task (e.g., predicting student performance). It is important that:

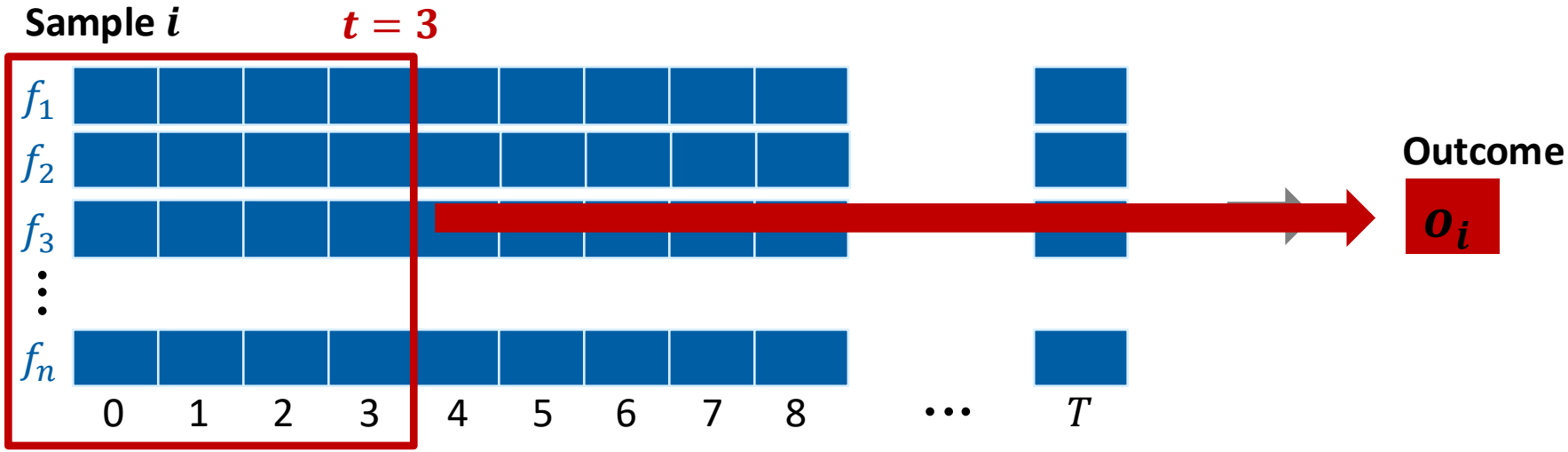
- You provide a clean and complete evaluation of your solution
- You are able to justify your decisions for each step

Agenda

- Model Evaluation:
  - Model Selection and Assessment
  - Error Metrics and Reporting Results
- **Time Series – Bayesian Knowledge Tracing**

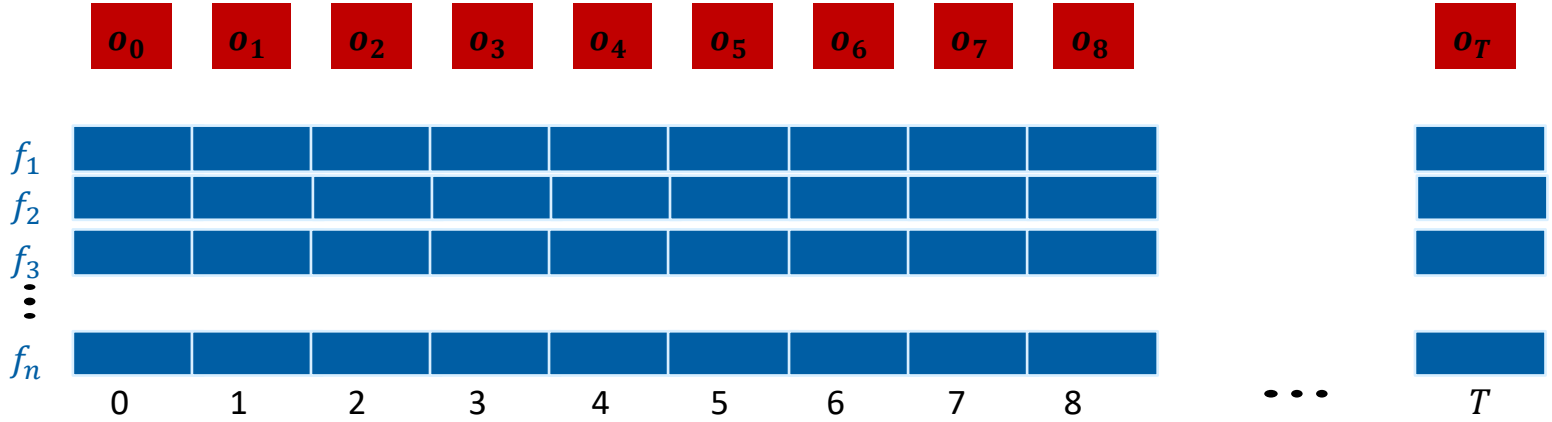
Time Series – Prediction Task

- Prediction of a target variable after $t < T$ time steps, where T is the total number of time steps



Time Series – Tracing Task

- Prediction of a variable in time step $t + 1$, based on time steps $0, \dots, t$



Handling Time Series Data – The naïve way

- Flattening
  - The number of parameters of the model depends on the number of time steps of the model
- Aggregation
  - Averaging across weeks
  - Accumulating across weeks

Today: Tracing Student Knowledge

- Is the student learning?
  - Measure what the student *knows* at a specific time *t*
  - More specifically: knowledge of the student about relevant knowledge components (skills)

Task:	$50 - 23 = ?$	$75 - 12 = ?$	$38 - 14 = ?$
Answer:	27	61	24

Tracing Knowledge – why is it useful?

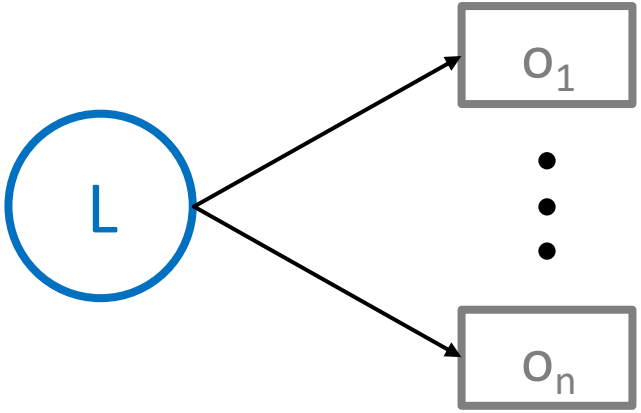
- Is the student learning?
  - Measure what the student *knows* at a specific time *t*
  - More specifically: knowledge of the student about relevant knowledge components (skills)

➡ Choose the next appropriate activity

➡ Know which activities support learning

What is a latent variable?

- A latent variable L is a variable which is not directly observable/cannot be measured
- It is assumed to affect the outcome of other variables $\boldsymbol{o}$, which can be observed (directly measured)

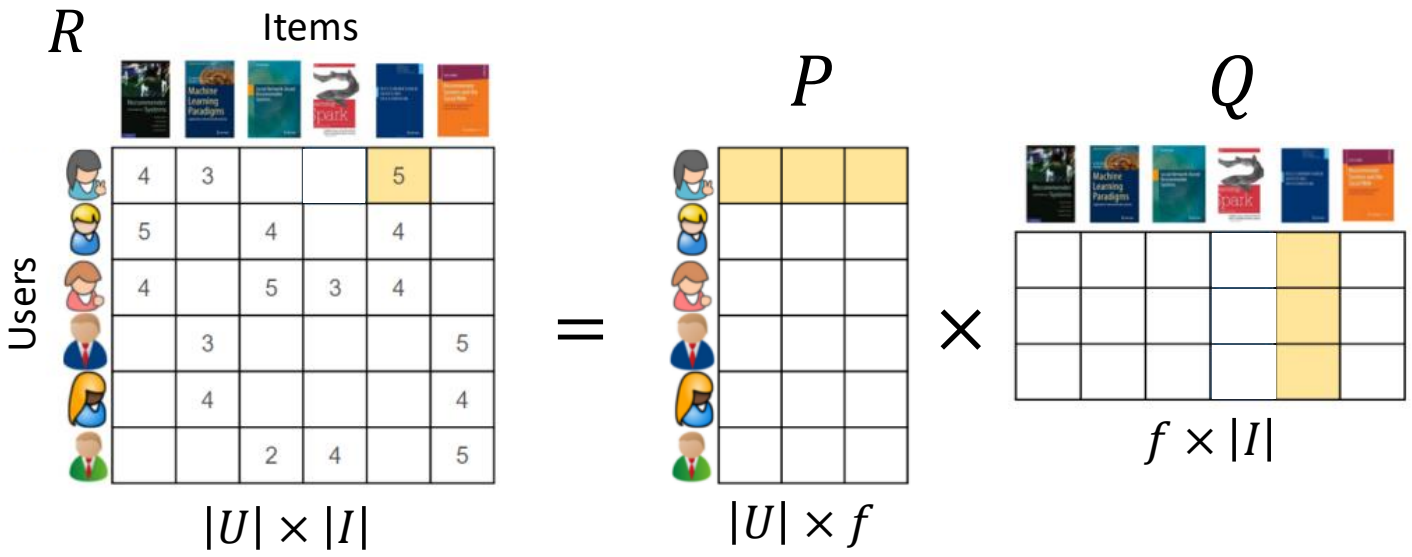


Why should we use latent variables?

- In many scientific fields, we are interested in concepts/factors that cannot directly be measured/observed:
  - Political sciences: leadership, political competence, etc.
  - Psychology: stress, self-worth, personality characteristics, talent, etc.
  - Education: memory, spatial ability, cognitive abilities, etc.
- We represent underlying concepts/factors by latent variables and infer them from the observed variables

Example 1: Recommender Systems

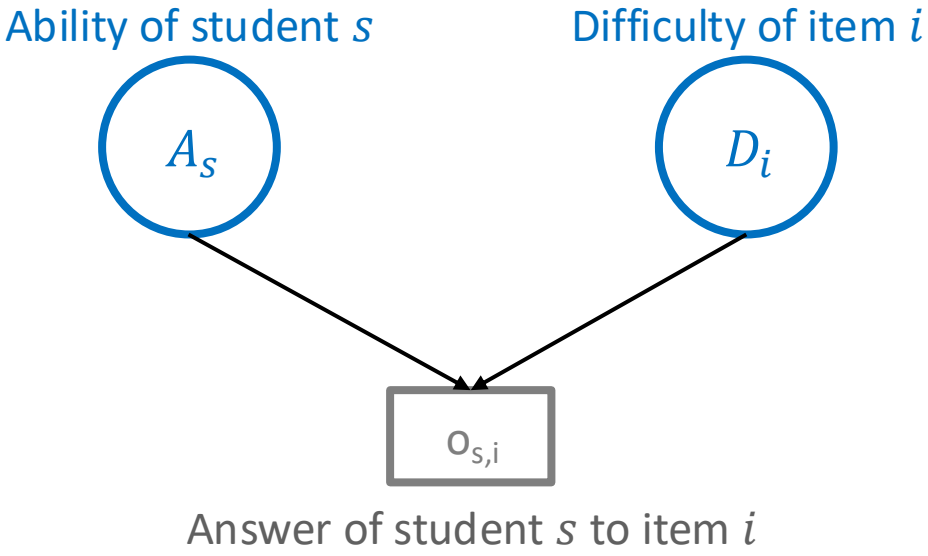
- Given: ratings of users u for items i (e.g., books)



[Image source: https://heartbeat.fritz.ai/recommender-systems-with-python-part-iii-collaborative-filtering-singular-value-decomposition-5b5dcb3f242b]

Example 2: Education

- Observations: binary answers (correct/wrong) of students to items (tasks)



Is the student learning?

Task:	$50 - 23 = ?$	$75 - 12 = ?$	$38 - 14 = ?$
Answer:	27	61	24

What are we measuring?

Task:	$50 - 23 = ?$	$75 - 12 = ?$	$38 - 14 = ?$
Answer:	27	61	24
	1	0	1

Binary observations of student answers

Subtraction 0-100

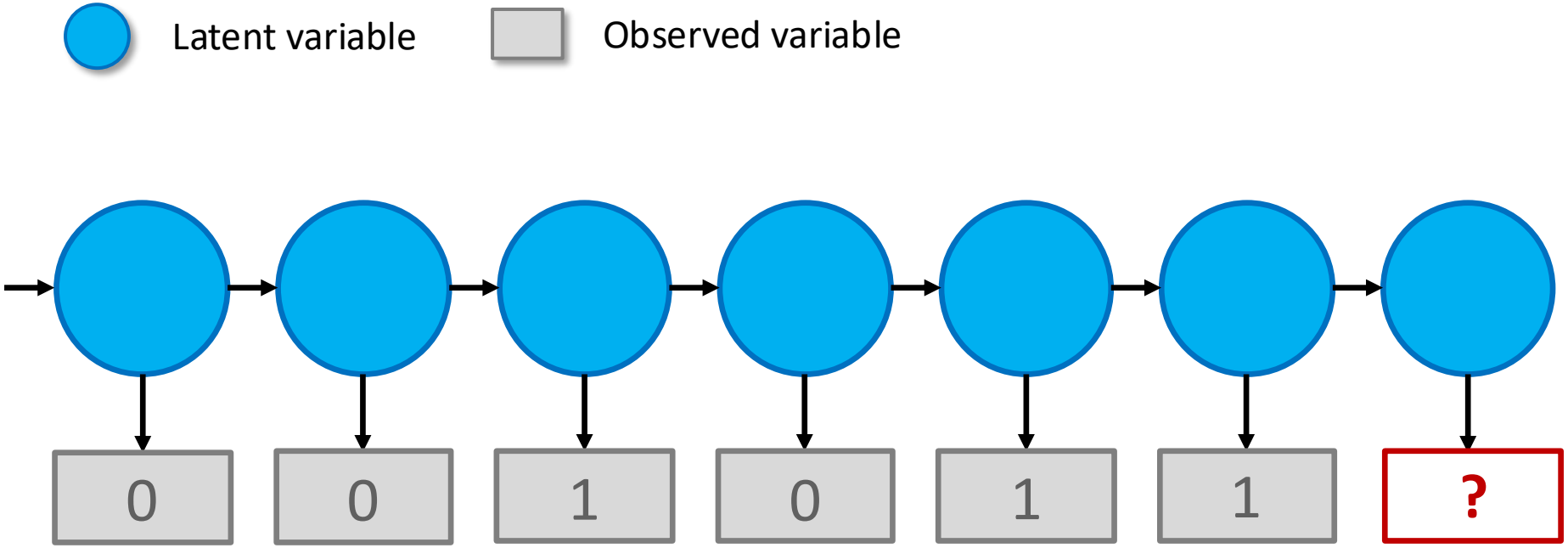
1	2		…		n
0	0	1	0	1	1

Predicting future performance

Subtraction 0-100

1	2		…		n	n+1
0	0	1	0	1	1	?

Bayesian Knowledge Tracing (BKT)

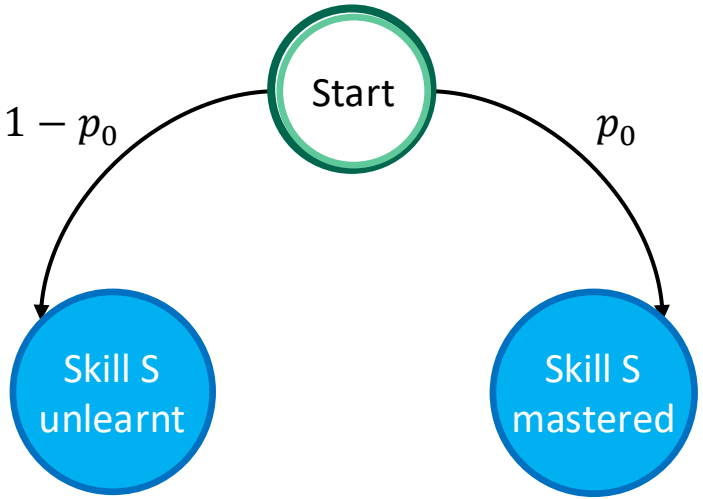


[Corbett & Anderson, UMUAI 1994]

Bayesian Knowledge Tracing (BKT)

Observations for student s:

Bayesian Knowledge Tracing (BKT)



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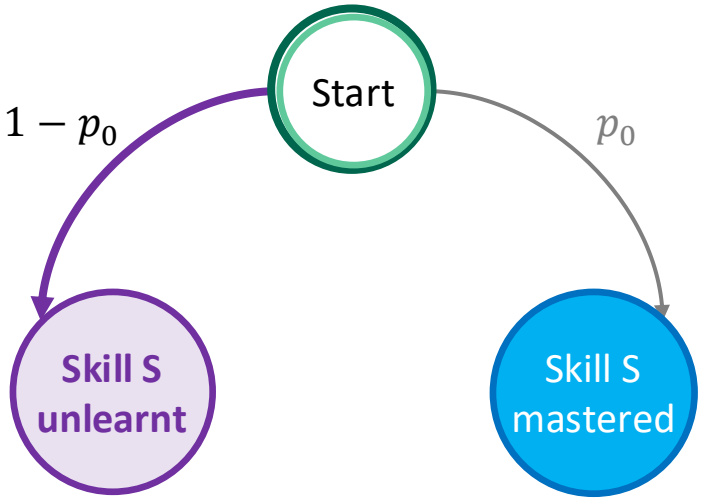
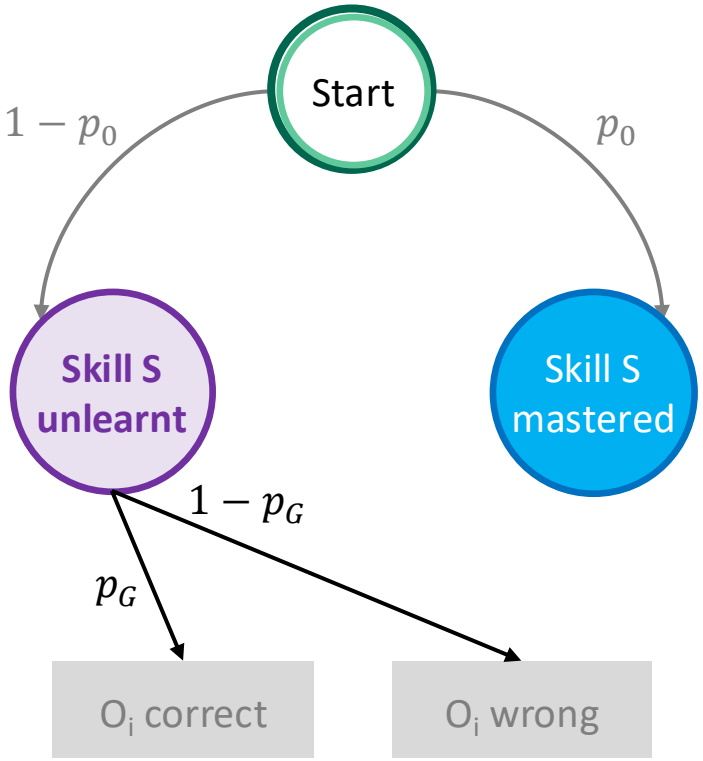


Observations for student s:

$t = 0$:

Bayesian Knowledge Tracing (BKT)



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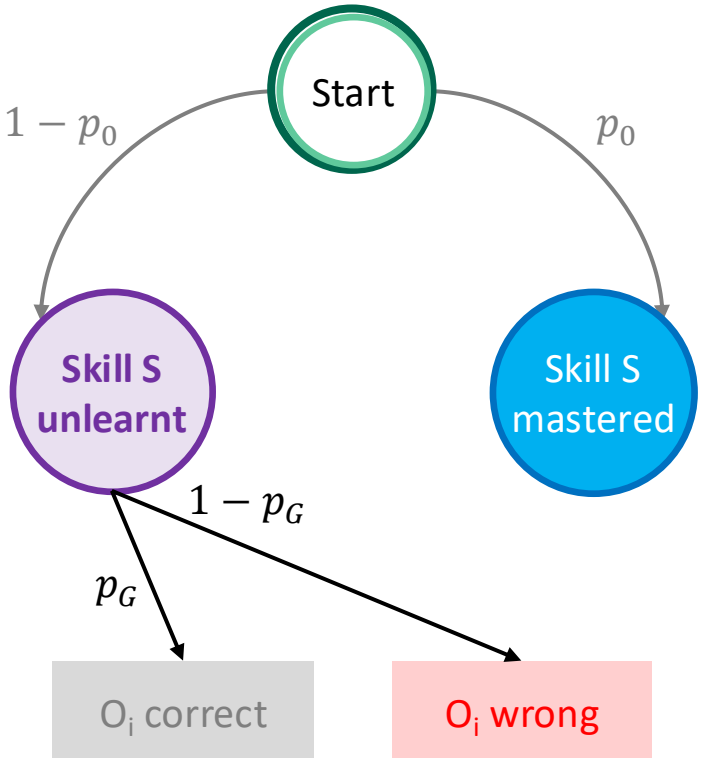
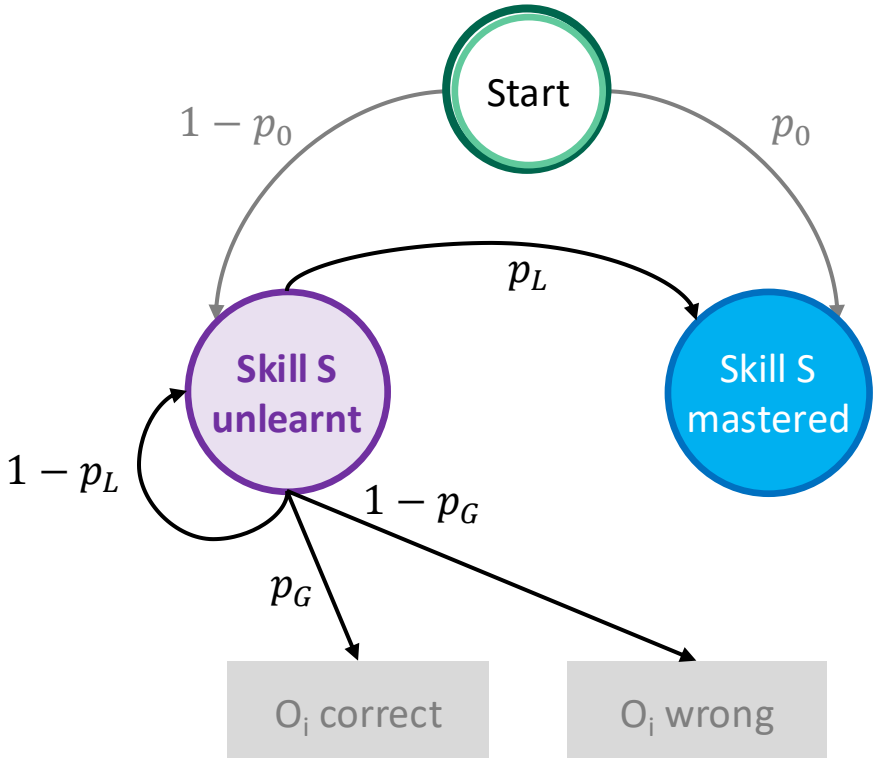


Observations for student s:

$t = 0$: 0

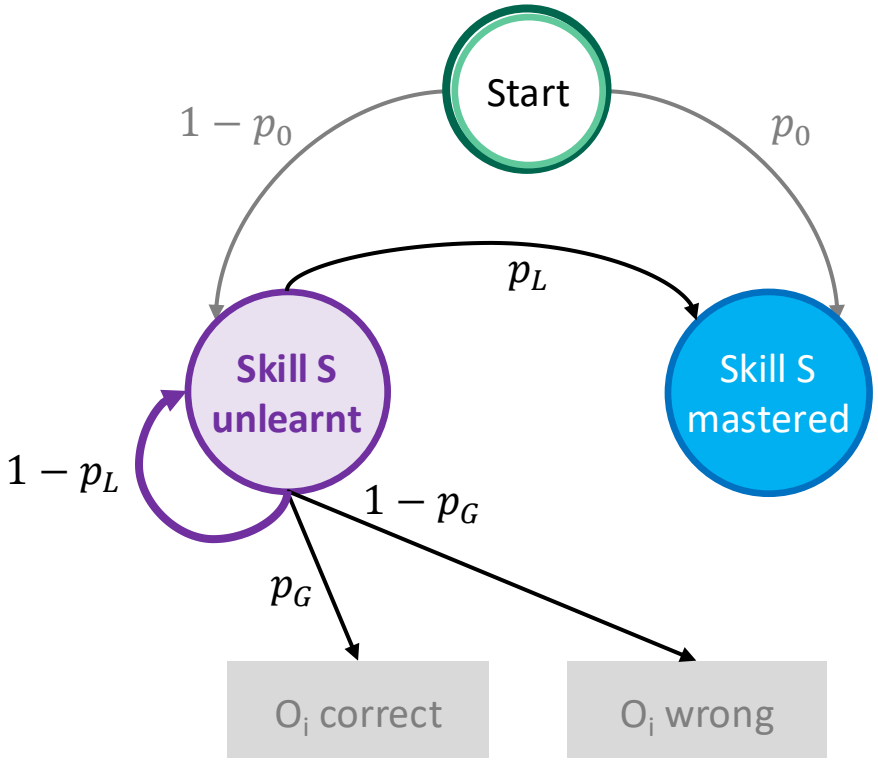
Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

Bayesian Knowledge Tracing (BKT)

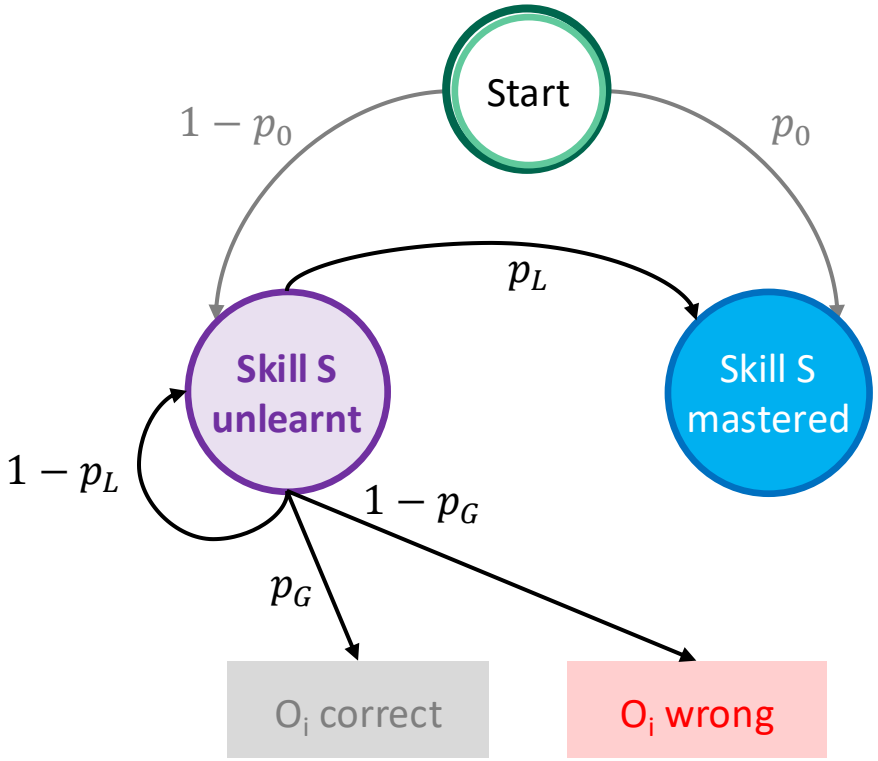


Observations for student s:

$t = 0$: 0

$t = 1$:

Bayesian Knowledge Tracing (BKT)

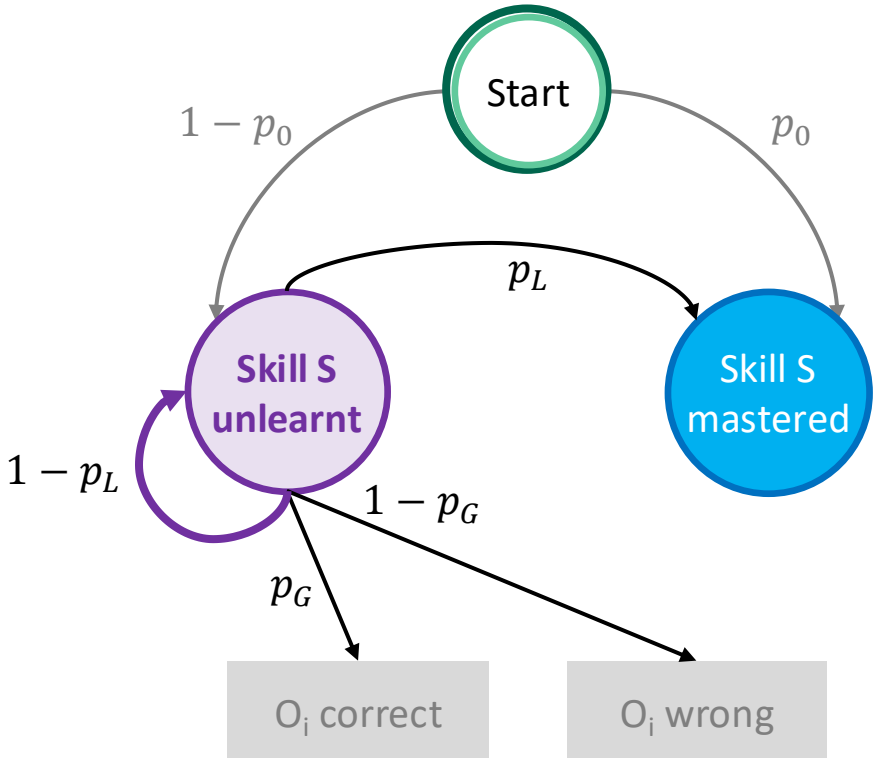


Observations for student s:

$t = 0$: 0

$t = 1$: 0

Bayesian Knowledge Tracing (BKT)



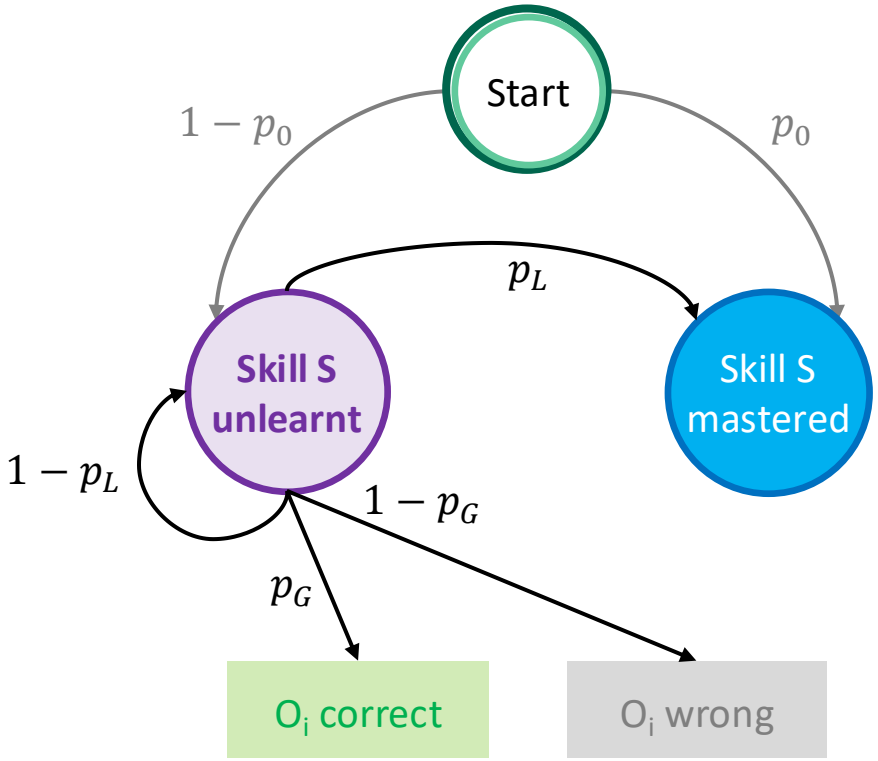
Observations for student s:

$t = 0$: 0

$t = 1$: 0

$t = 2$:

Bayesian Knowledge Tracing (BKT)



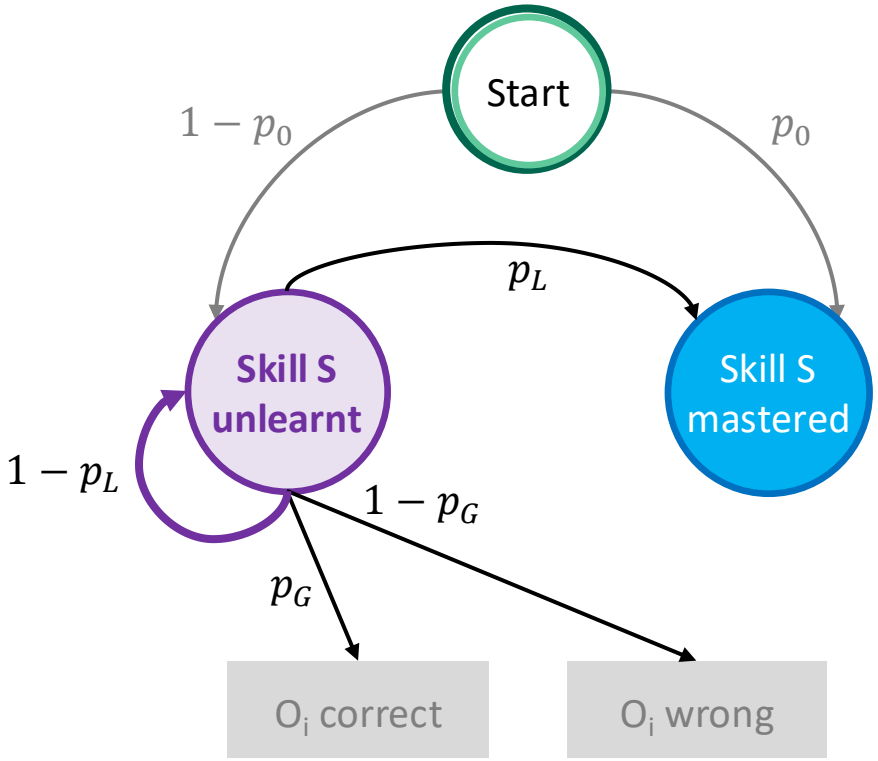
Observations for student s:

$t = 0$: 0

$t = 1$: 0

$t = 2$: 1

Bayesian Knowledge Tracing (BKT)



Observations for student s:

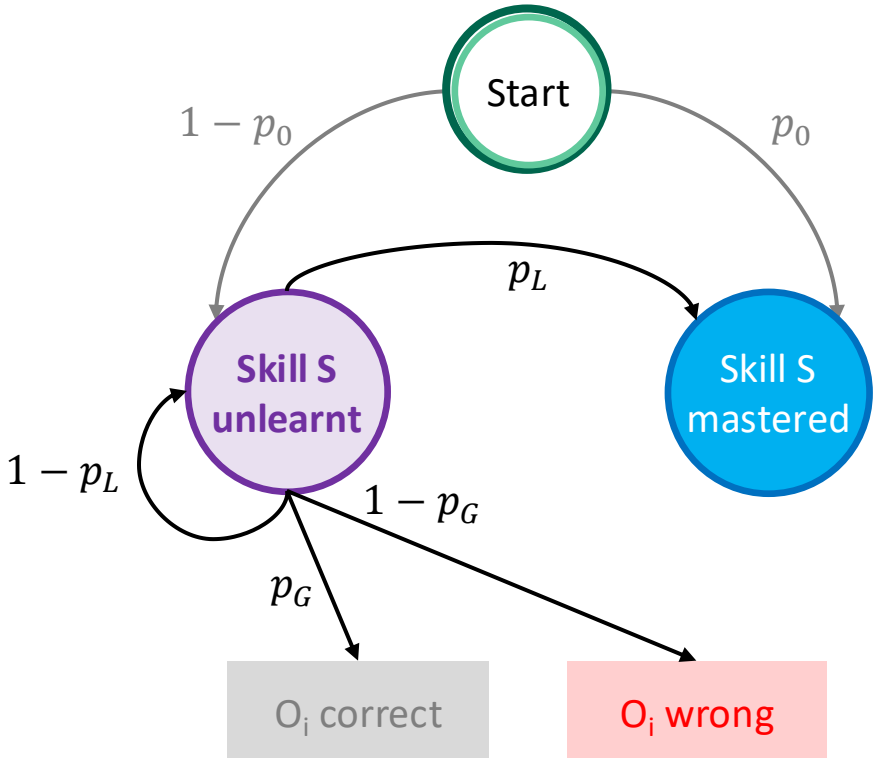
$t = 0$: 0

$t = 1$: 0

$t = 2$: 1

$t = 3$:

Bayesian Knowledge Tracing (BKT)



Observations for student s:

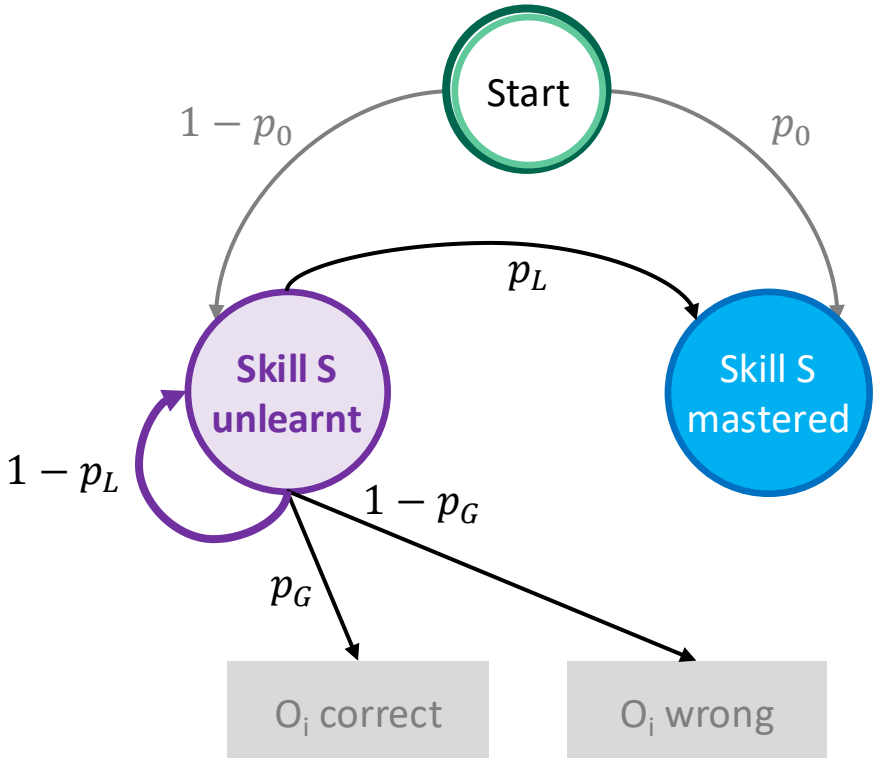
$t = 0$: 0

$t = 1$: 0

$t = 2$: 1

$t = 3$: 0

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

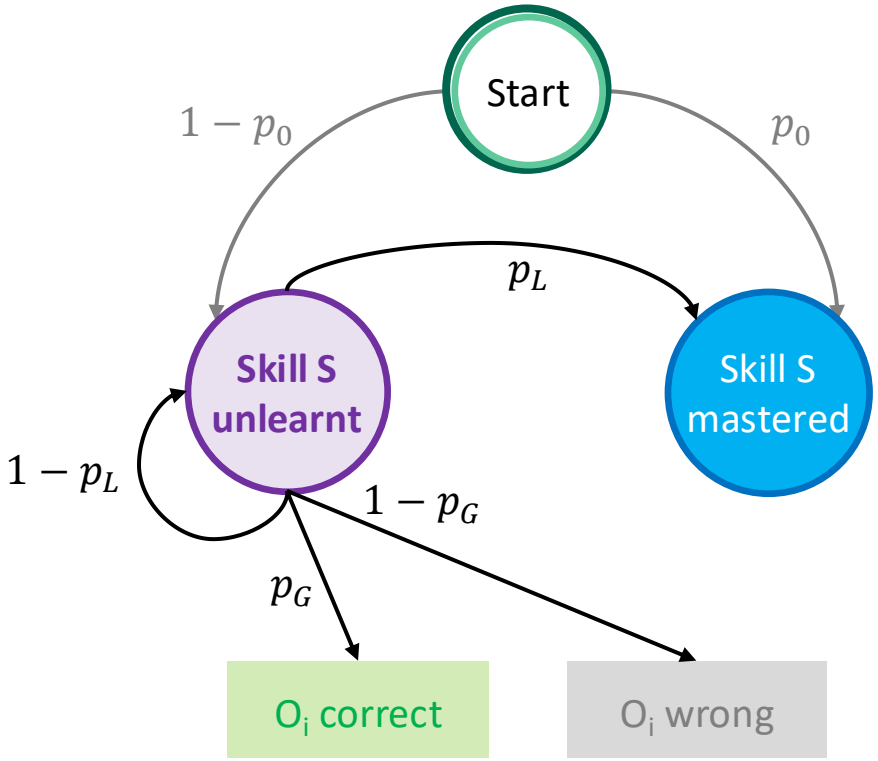
$t = 1$: 0

$t = 2$: 1

$t = 3$: 0

$t = 4$:

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

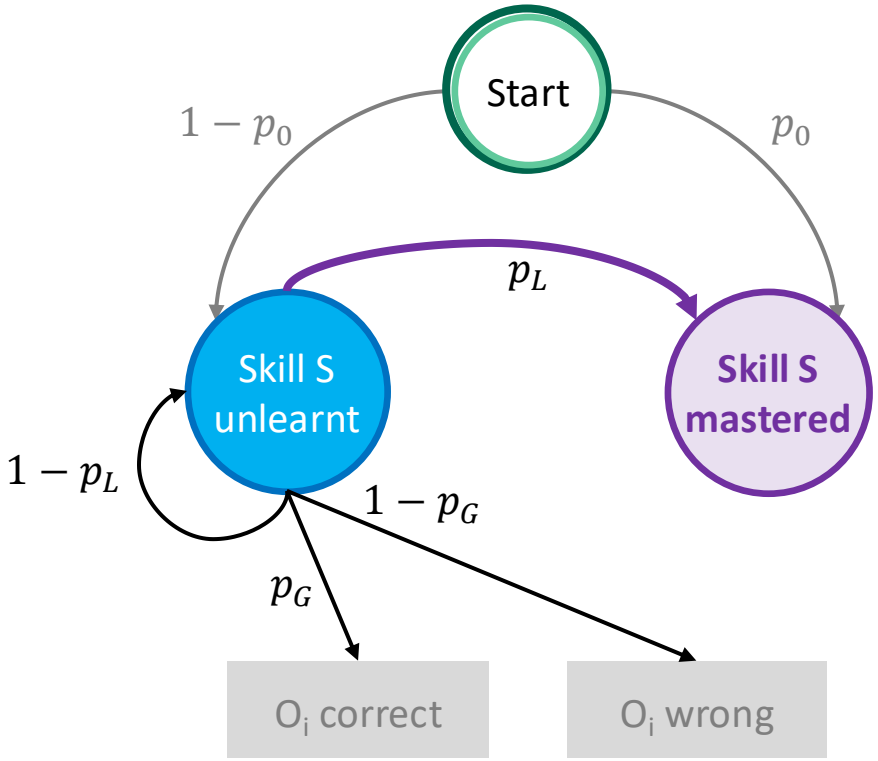
$t = 1$: 0

$t = 2$: 1

$t = 3$: 0

$t = 4$: 1

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

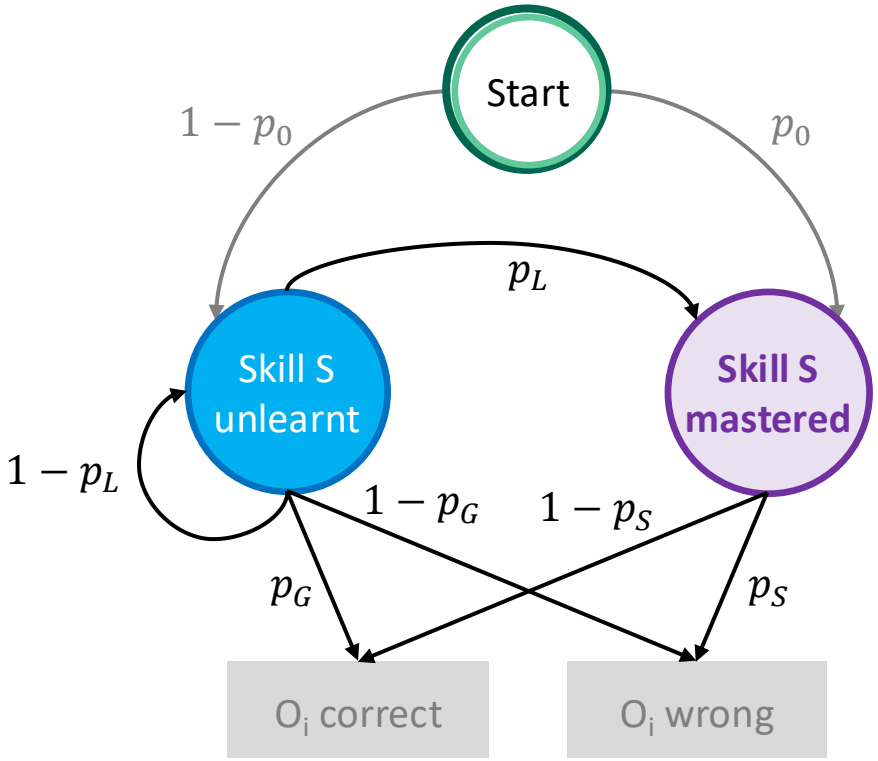
$t = 2$: 1

$t = 3$: 0

$t = 4$: 1

$t = 5$:

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

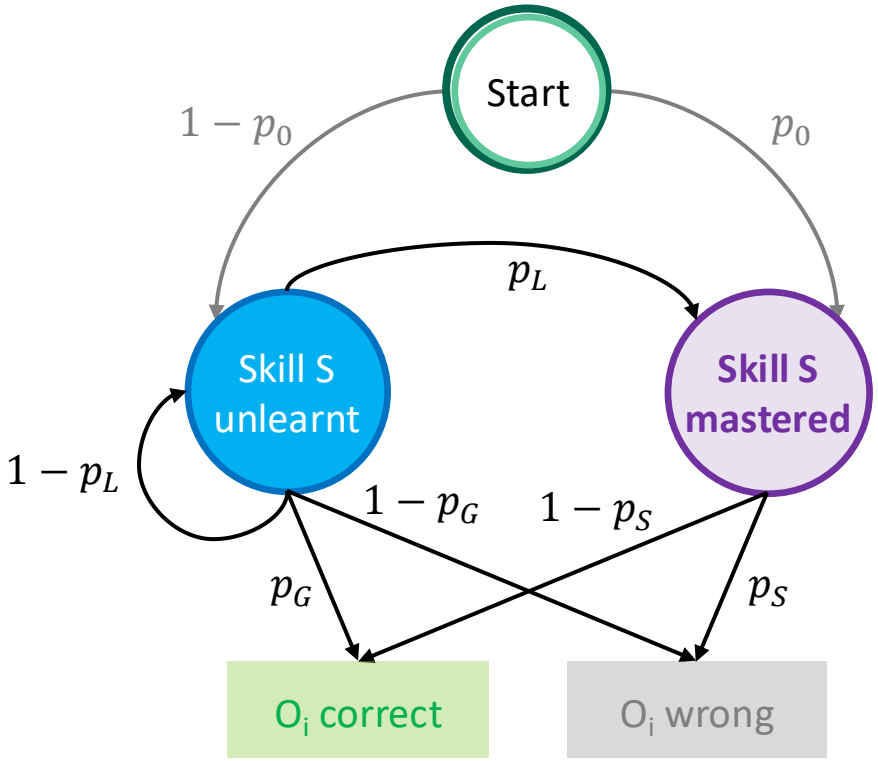
$t = 2$: 1

$t = 3$: 0

$t = 4$: 1

$t = 5$:

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

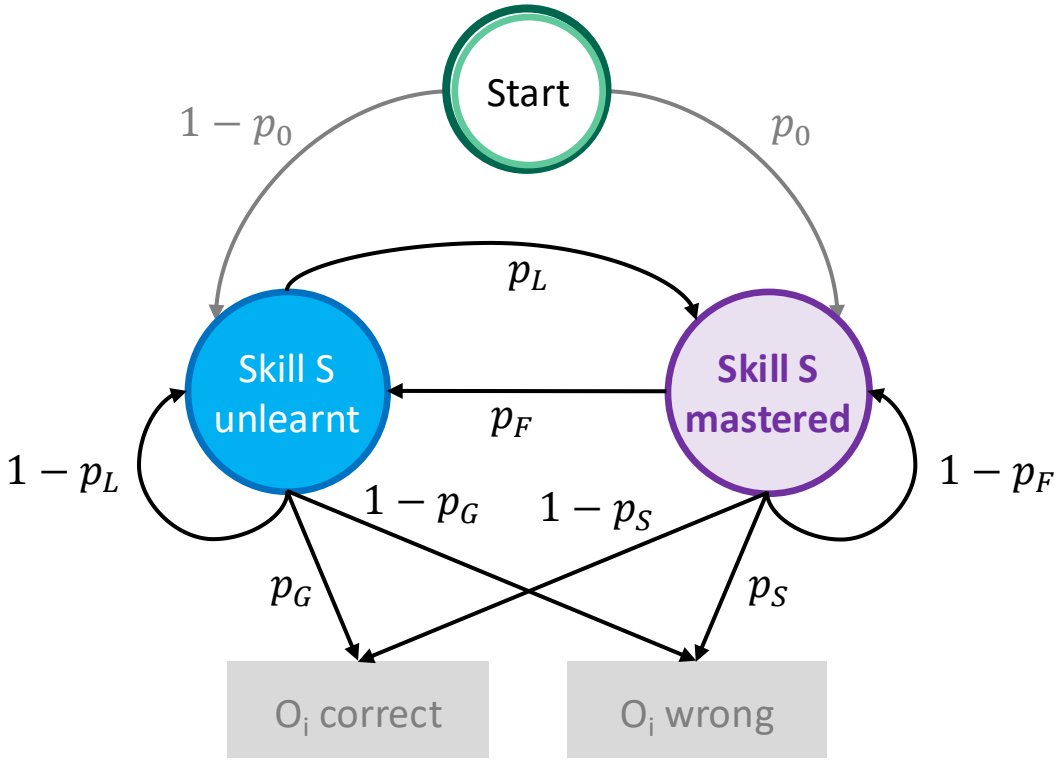
$t = 2$: 1

$t = 3$: 0

$t = 4$: 1

$t = 5$: 1

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

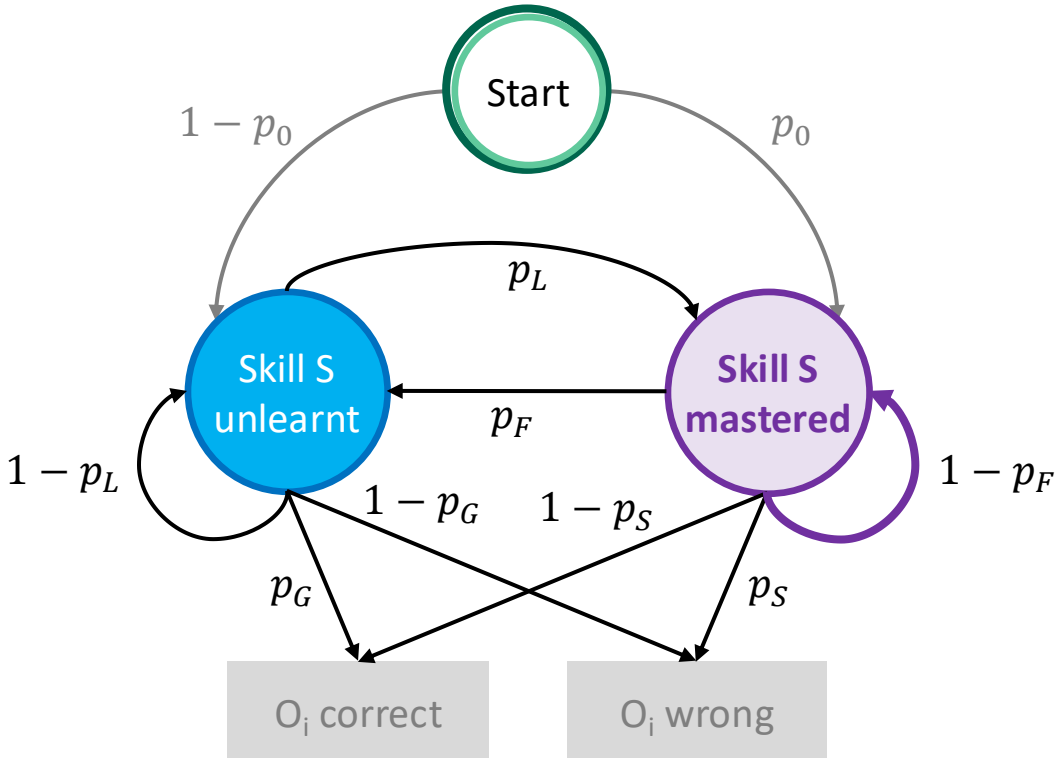
$t = 2$: 1

$t = 3$: 0

$t = 4$: 1

$t = 5$: 1

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

$t = 2$: 1

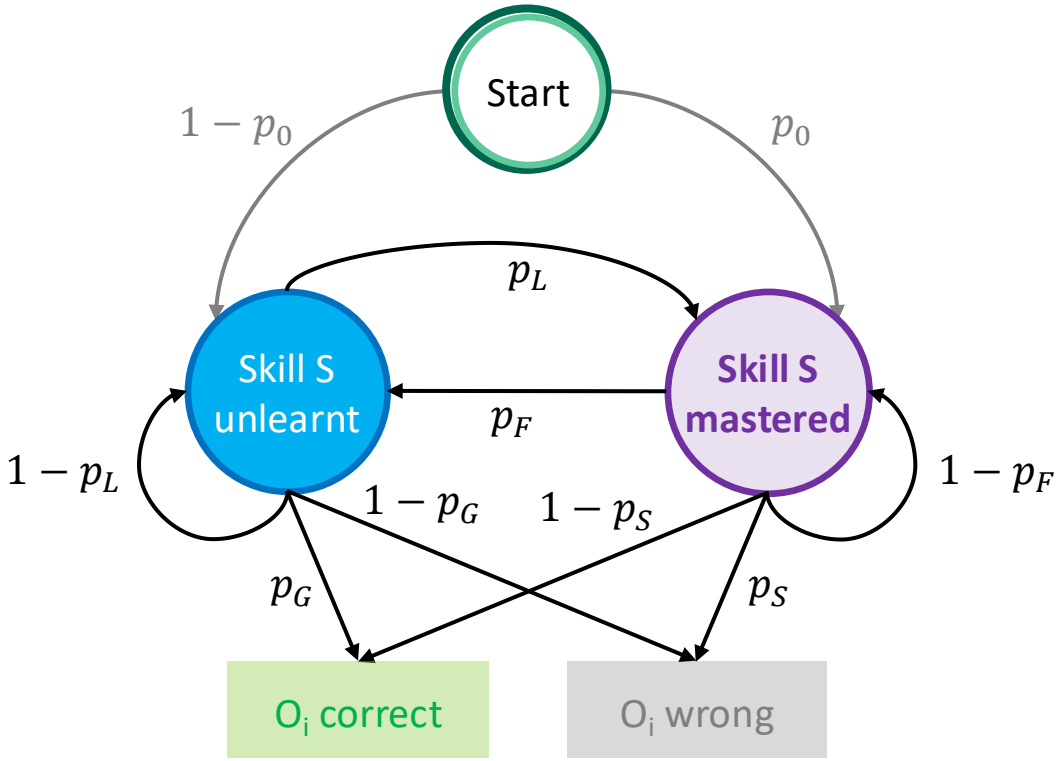
$t = 3$: 0

$t = 4$: 1

$t = 5$: 1

$t = 6$:

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

$t = 2$: 1

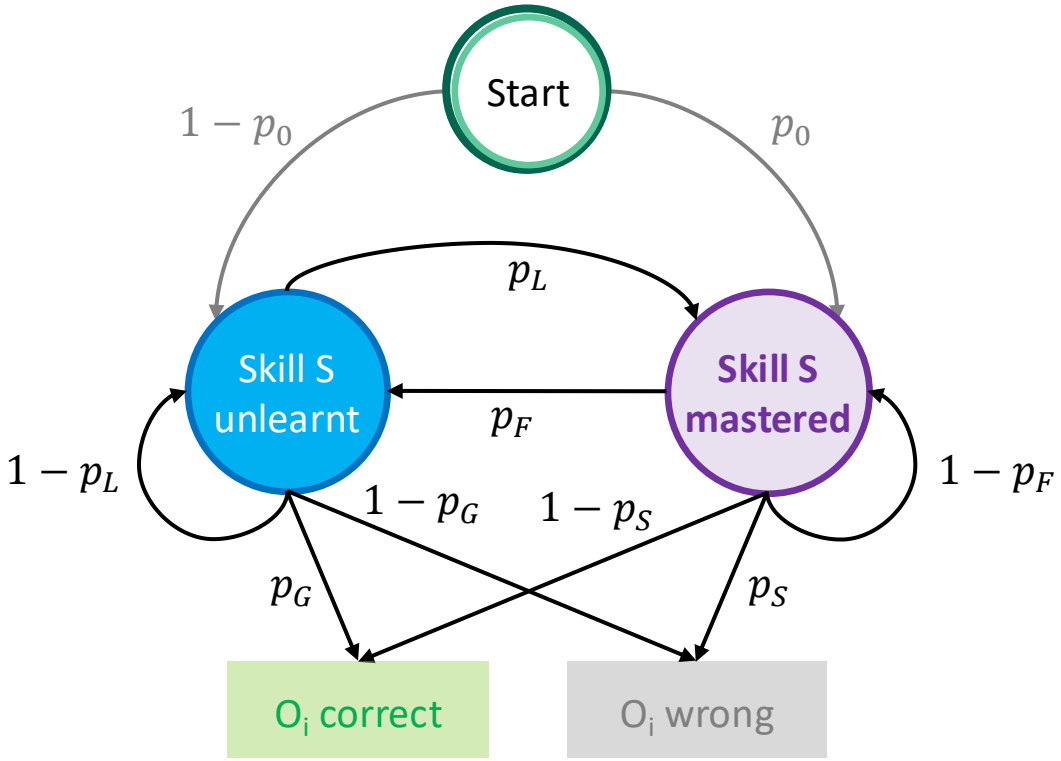
$t = 3$: 0

$t = 4$: 1

$t = 5$: 1

$t = 6$: 1

Bayesian Knowledge Tracing (BKT)



Observations for student s:

$t = 0$: 0

$t = 1$: 0

$t = 2$: 1

$t = 3$: 0

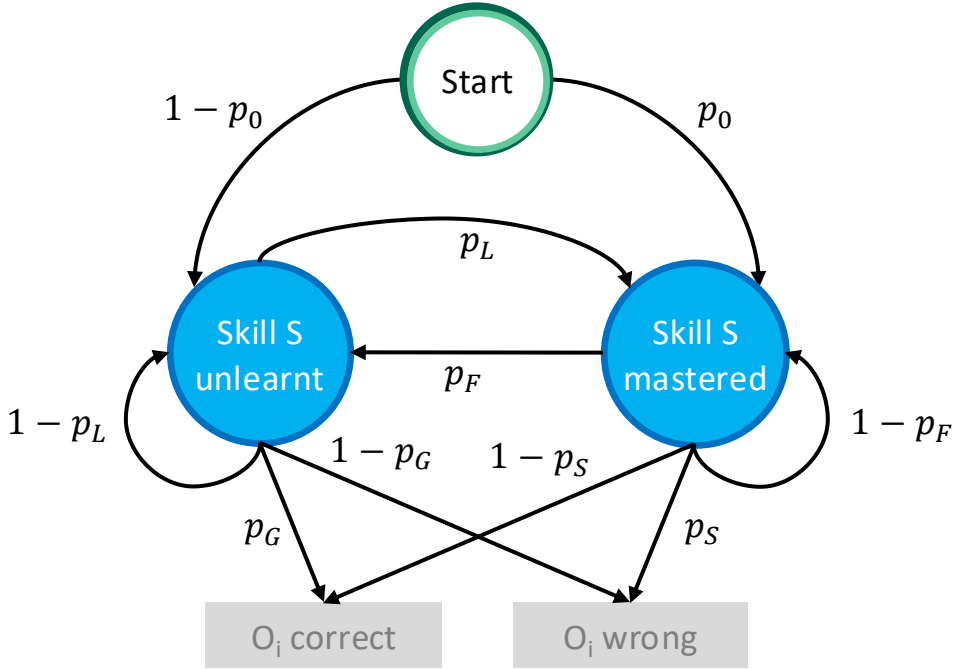
$t = 4$: 1

$t = 5$: 1

$t = 6$: 1

$$\boldsymbol{o_s} = [0,0,1,0,1,1,1]$$

BKT - Terminology



One latent variable (S) with two possible states

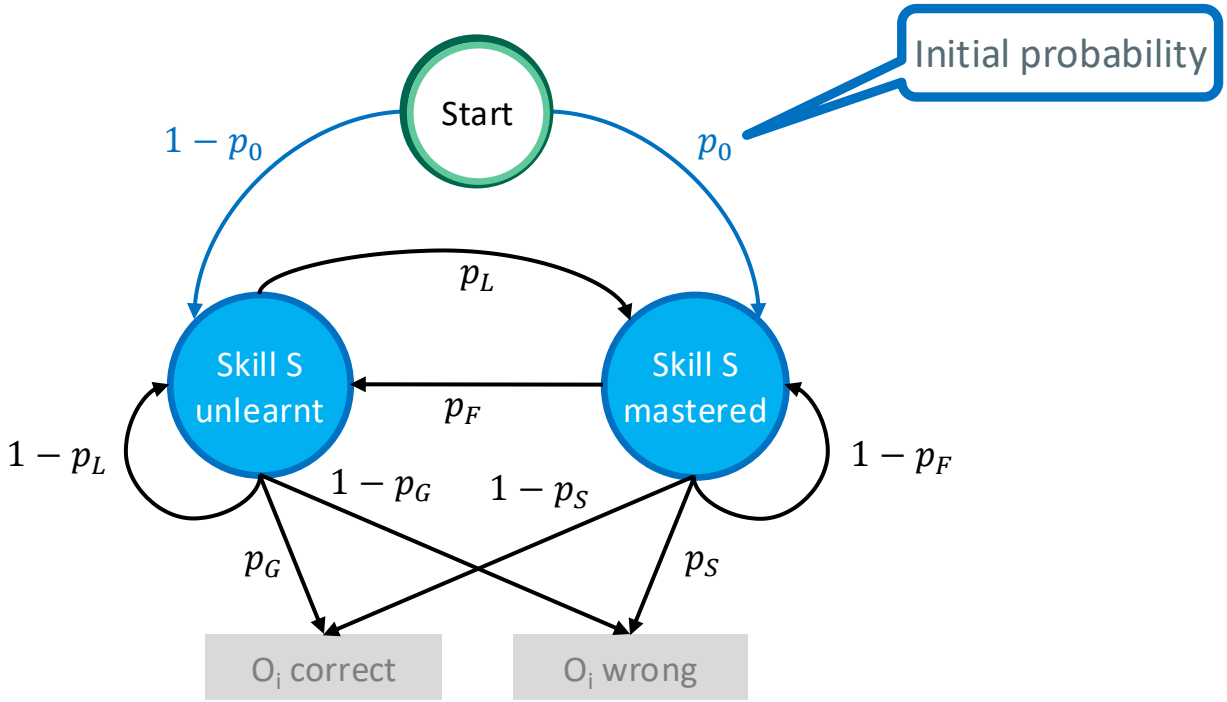
Observations (also binary)

Five parameters:

$$\theta = \{p_0, \underbrace{p_L, p_F}_{\text{Transition probabilities}}, \overbrace{p_S, p_G}^{\text{Emission probabilities}}\}$$

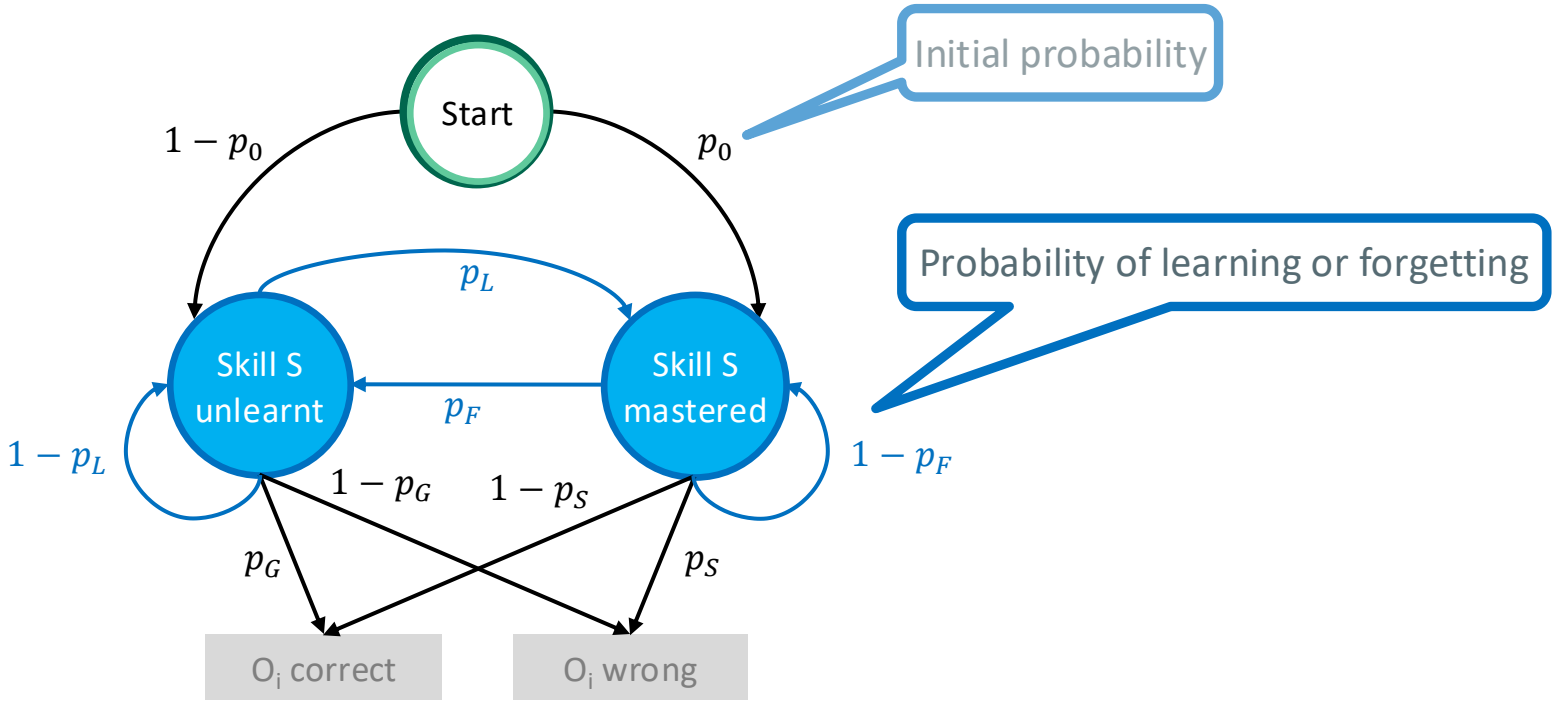
[Corbett & Anderson, UMUAI 1994]

BKT parameters are interpretable



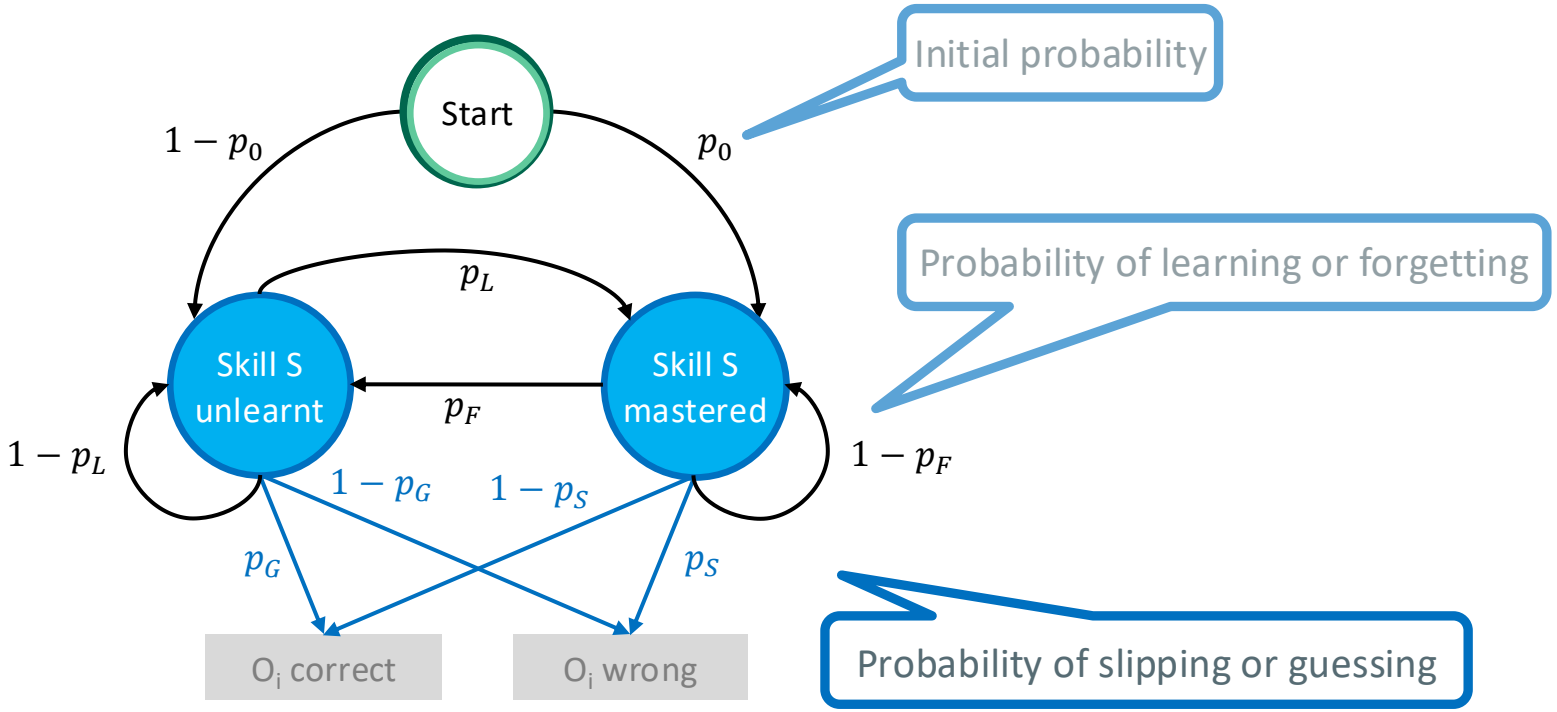
[Corbett & Anderson, UMUAI 1994]

BKT parameters are interpretable



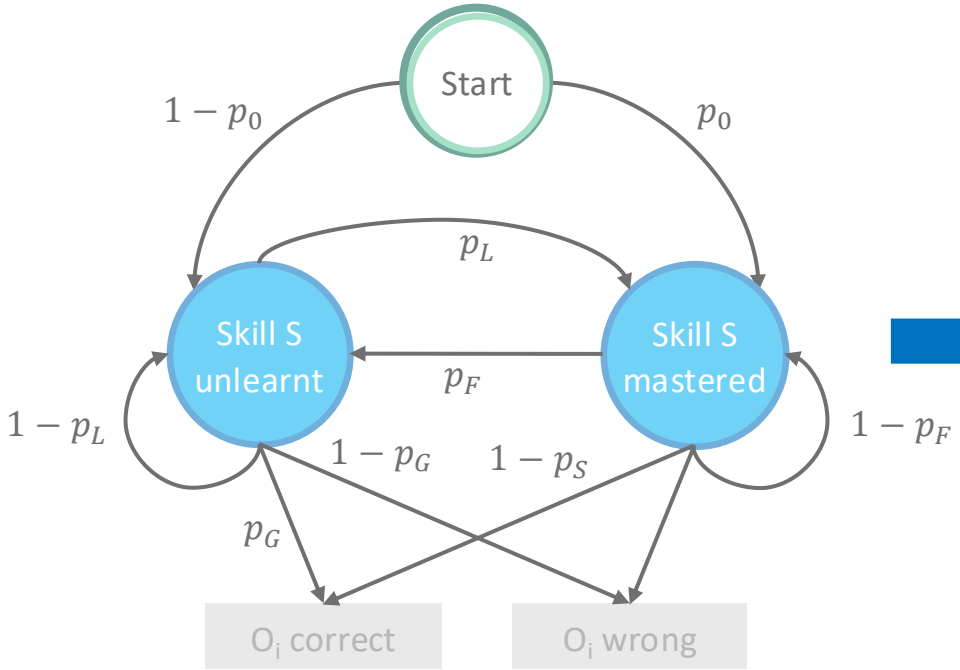
[Corbett & Anderson, UMUAI 1994]

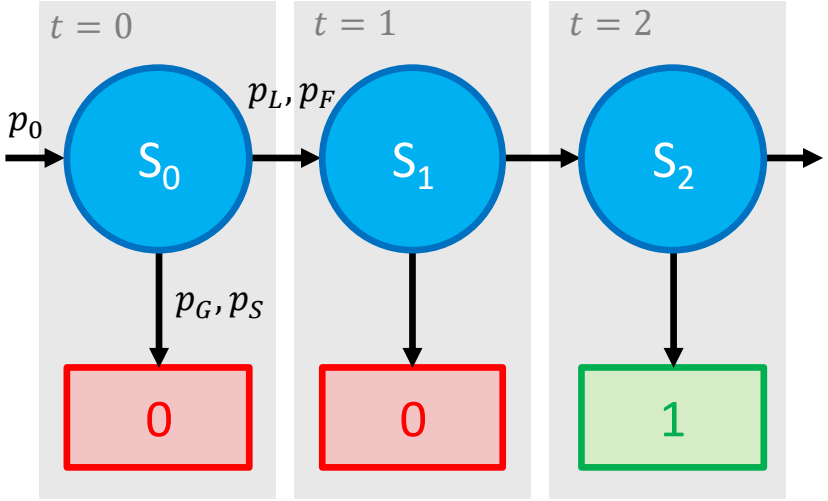
BKT parameters are interpretable



[Corbett & Anderson, UMUAI 1994]

BKT – unrolled over time





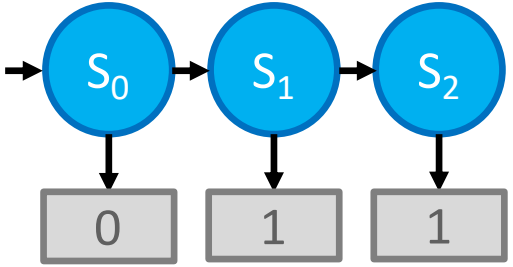
Two tasks need to be solved in practice

- Given a model with parameters $\theta = \{p_0, p_L, p_F, p_S, p_G\}$ and a sequence of observations $\boldsymbol{o} = [o_0, \dots, o_t]$ from a student s, predict o_{t+1}

o_0	o_1	o_2	$\cdots$		o_t	o_{t+1}
0	0	1	0	1	1	?

Inference

Inference Example



$$p_0 = 0.5$$
$$p_S = 0.2$$
$$p_G = 0.3$$
$$p_L = 0.4$$
$$p_F = 0.0$$

S_0	$p(S_0)$
1	p_0
0	$1-p_0$

S_t	S_{t+1}	$p(S_{t+1}\|S_t)$
0	0	$1-p_L$
0	1	p_L
1	0	p_F
1	1	$1-p_F$

S_t	O_t	$p(O_t\|S_t)$
0	0	$1-p_G$
0	1	p_G
1	0	p_S
1	1	$1-p_S$

Some useful rules:

$$p(A,B) = p(A|B) \cdot p(B)$$

$$p(A|B) = \frac{p(B|A) \cdot p(A)}{p(B)}$$

$$p(A=1) = p(A=1, B=1) + p(A=1, B=0)$$

- $p(s_0 = 1)$?
- $p(o_0 = 1)$?
- $p(s_1 = 1|o_0 = 0)$?
- $p(o_1 = 1|o_0 = 0)$?
- $p(s_2 = 1|o_0 = 0, o_1 = 1)$?

Inference in BKT models

$\boldsymbol{o_{t-1}} = [o_0, \dots, o_{t-1}]$

Equations for time $t = 0$:

Belief about latent state before observation

$$p(s_0 = 1) = p_0$$

Predicted observation at time t

$$p(o_0 = 1) = (1 - p_s) \cdot p_0 + p_G \cdot (1 - p_0)$$

$$p(o_0 = 0) = p_S \cdot p_0 + (1 - p_G) \cdot (1 - p_0)$$

Posterior: belief about latent state after observation

$$\left.\begin{aligned} p_{s_0|1} &= \frac{(1 - p_S) \cdot p_0}{(1 - p_s) \cdot p_0 + p_G \cdot (1 - p_0)} \\ p_{s_0|0} &= \frac{p_S \cdot p_0}{p_s \cdot p_0 + (1 - p_G) \cdot (1 - p_0)} \end{aligned}\right\} p_{s_0|o_0}$$

Equations for time steps $t = 1, \dots, T$:

$$p_{s_t|\boldsymbol{o_{t-1}}} = (1 - p_F) \cdot p_{s_{t-1}|\boldsymbol{o_{t-1}}} + p_L \cdot (1 - p_{s_{t-1}|\boldsymbol{o_{t-1}}})$$

$$p(o_t = 1|\boldsymbol{o_{t-1}}) = (1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}} + p_G \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})$$

$$p(o_t = 0|\boldsymbol{o_{t-1}}) = p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}} + (1 - p_G) \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})$$

$$\left.\begin{aligned} p_{s_t|1,\boldsymbol{o_{t-1}}} &= \frac{(1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}}}{(1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}} + p_G \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})} \\ p_{s_t|0,\boldsymbol{o_{t-1}}} &= \frac{p_s \cdot p_{s_t|\boldsymbol{o_{t-1}}}}{p_s \cdot p_{s_t|\boldsymbol{o_{t-1}}} + (1 - p_G) \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})} \end{aligned}\right\} p_{s_t|\boldsymbol{o_t}}$$

[Corbett & Anderson, UMUAI 1994]

Making predictions using a BKT model

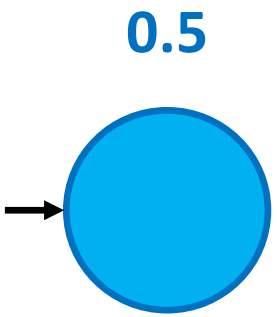
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



$p(s_0 = 1) = p_0$

Making predictions using a BKT model

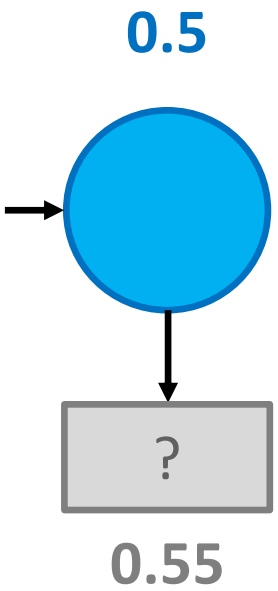
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



$$p(s_0 = 1) = p_0$$

$$p(o_0 = 1) = (1 - p_s) \cdot p_0 + p_G \cdot (1 - p_0)$$

Making predictions using a BKT model

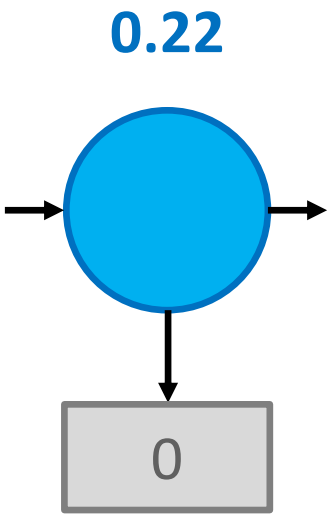
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



$$p_{s_0|0} = \frac{p_S \cdot p_0}{1 - p(o_0 = 1)} = \frac{0.2 \cdot 0.5}{0.45} = 0.22$$

Making predictions using a BKT model

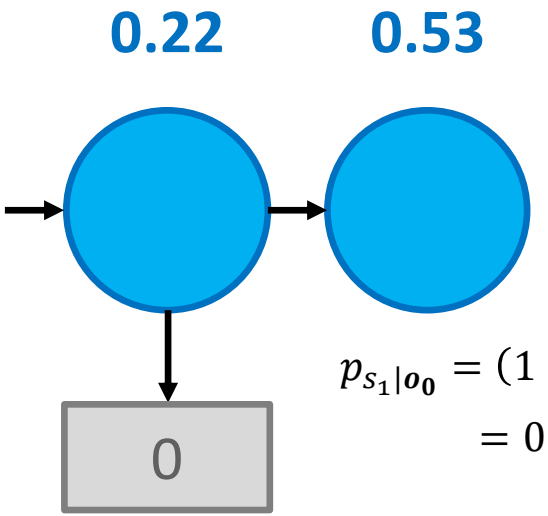
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



$$p_{s_1|\boldsymbol{o_0}} = (1 - p_F) \cdot p_{s_0|\boldsymbol{o_0}} + p_L \cdot (1 - p_{s_0|\boldsymbol{o_0}})$$
$$= 0.22 + 0.4 \cdot 0.78 = 0.53$$

Making predictions using a BKT model

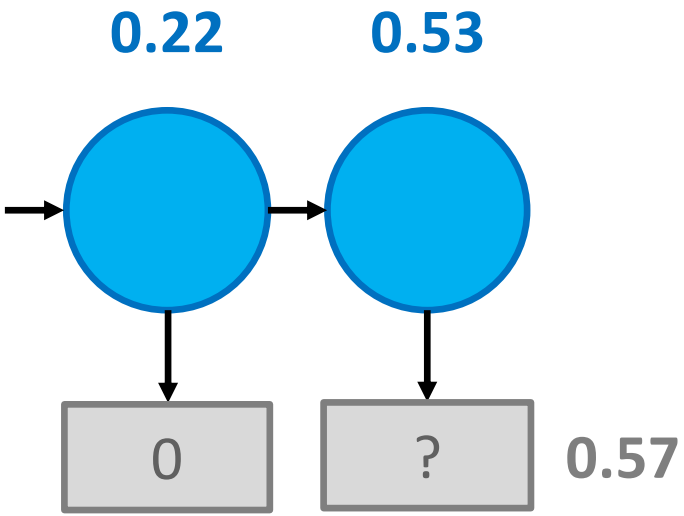
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



$$p(o_1 = 1|\boldsymbol{o_0}) = (1 - p_S) \cdot p_{s_1|\boldsymbol{o_0}} + p_G \cdot (1 - p_{s_1|\boldsymbol{o_0}})$$
$$= 0.8 \cdot 0.53 + 0.3 \cdot 0.47 = 0.57$$

Making predictions using a BKT model

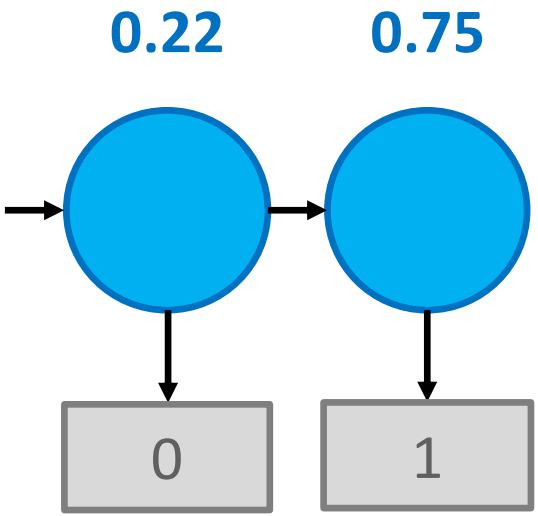
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



$$p_{s_1|1,\boldsymbol{o_0}} = \frac{(1-p_S)\cdot p_{s_1|\boldsymbol{o_0}}}{(1-p_S)\cdot p_{s_1|\boldsymbol{o_0}} + p_G\cdot(1-p_{s_1|\boldsymbol{o_0}})}$$

$$= \frac{0.8\cdot 53}{0.57} = 0.75$$

Making predictions using a BKT model

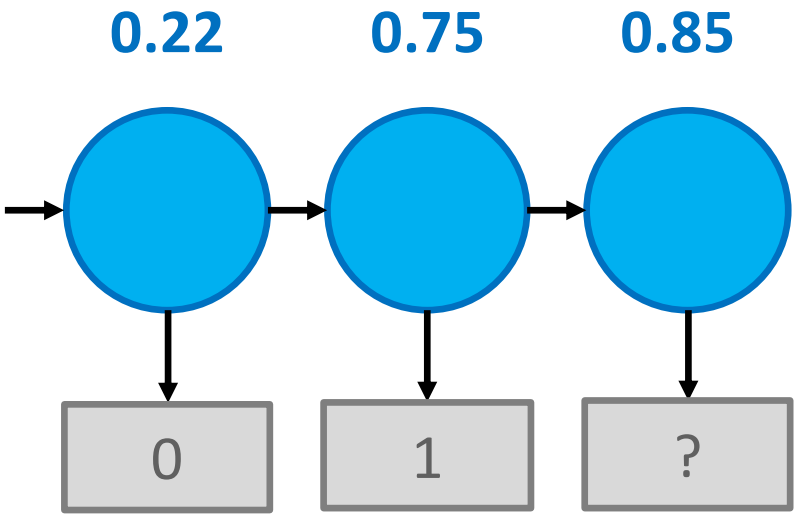
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



Making predictions using a BKT model

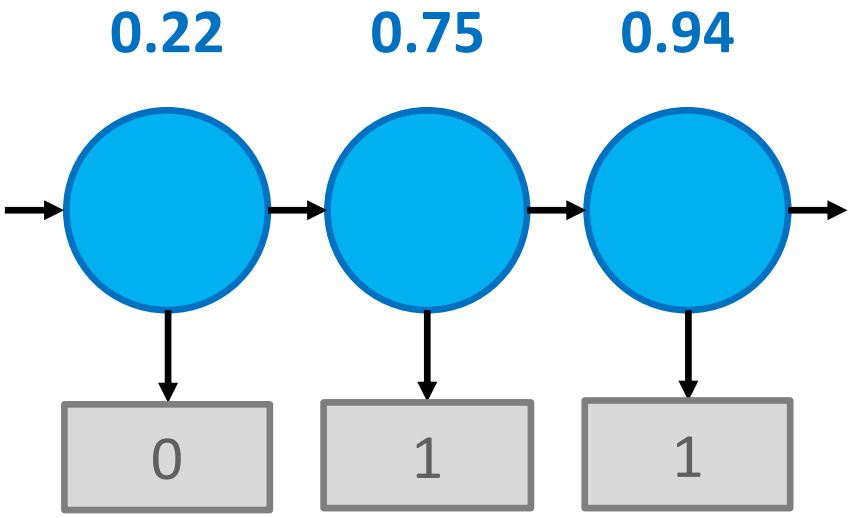
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



Making predictions using a BKT model

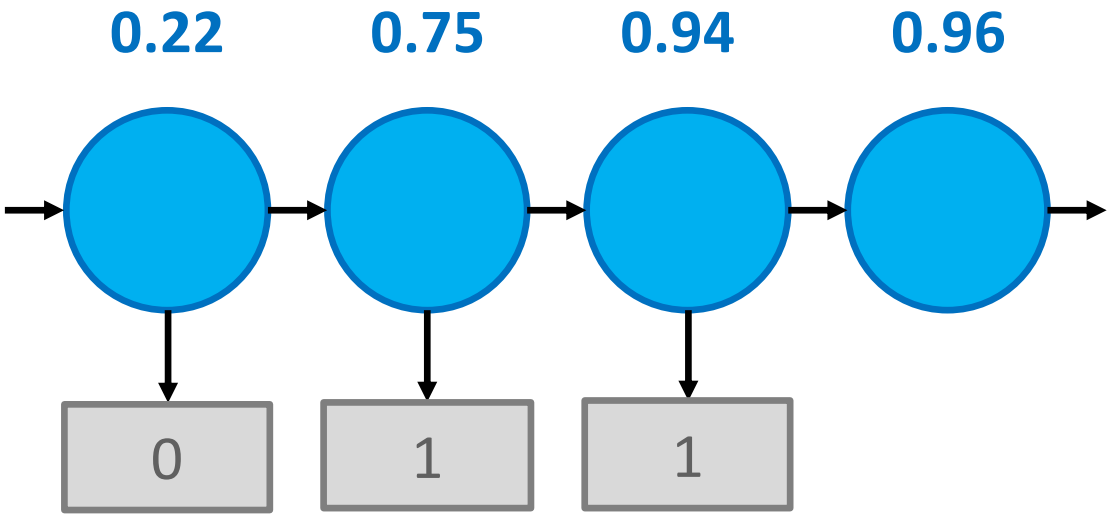
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



Making predictions using a BKT model

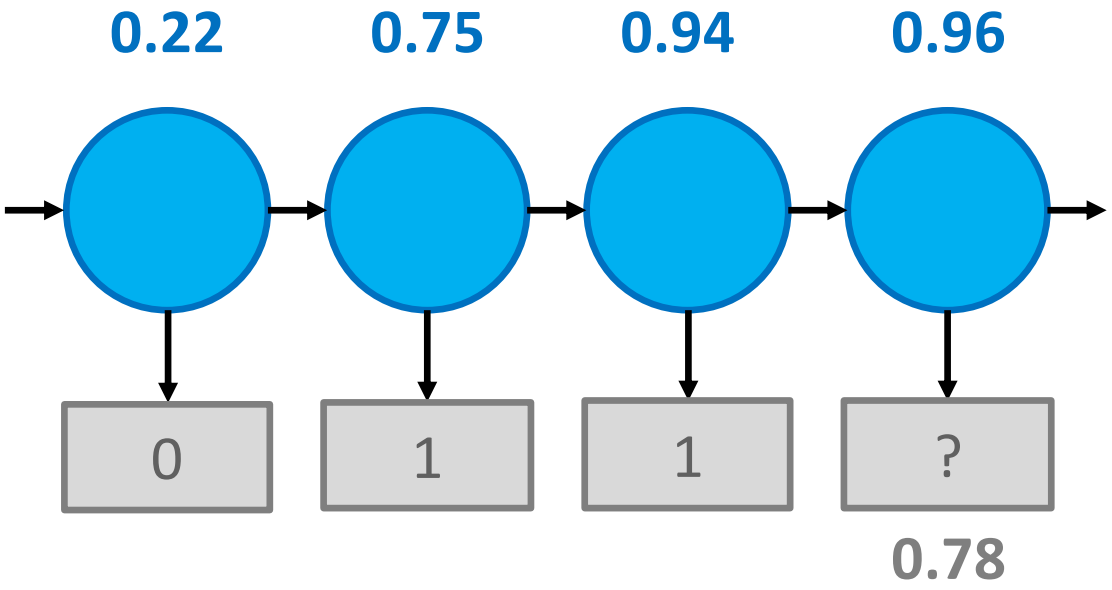
$p_0 = 0.5$

$p_S = 0.2$

$p_G = 0.3$

$p_L = 0.4$

$p_F = 0.0$



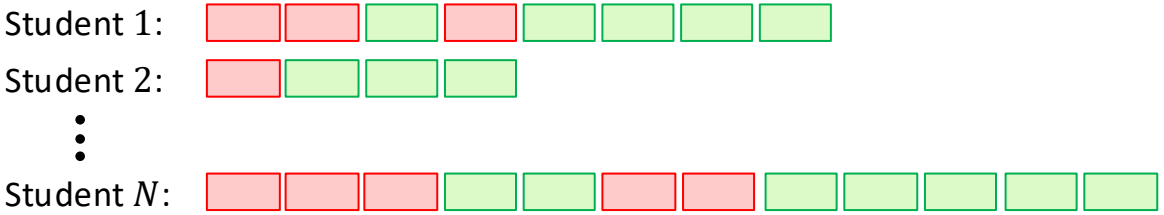
Two tasks need to be solved in practice

- Given a model with parameters $\theta = \{p_0, p_L, p_F, p_S, p_G\}$ and a sequence of observations $\boldsymbol{o} = [o_0, \dots, o_t]$ from a student s, predict o_{t+1}

o_0	o_1	o_2	$\cdots$		o_t	o_{t+1}
0	0	1	0	1	1	?

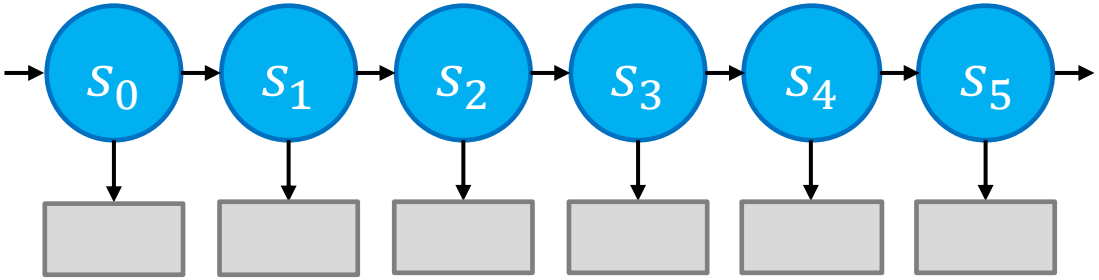
Inference

- Given sequences of observations $\boldsymbol{o} = [o_0, \dots, o_T]$ of N students, learn the parameters $\theta = \{p_0, p_L, p_F, p_S, p_G\}$ that maximize the likelihood of the observed data



Parameter Learning

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

Student l_0: $\mathbf{o}_{l_0} = [0,1,1]$

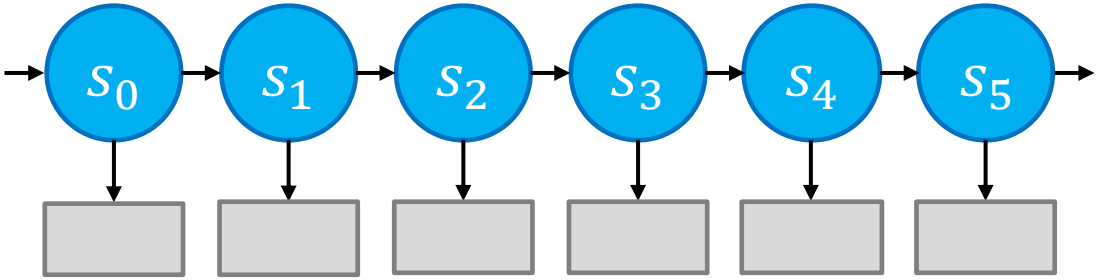
$\vdots$

Student l_{N-1}: $\mathbf{o}_{l_{N-1}} = [1,0,1,1,1,0,0,1,1,1]$

Student l_N: $\mathbf{o}_{l_N} = [0,1,0,1]$

$$\max_{\theta} \; p\left(\boldsymbol{o}_{l_0}, \dots, \mathbf{o}_{l_{N-1}}, \mathbf{o}_{l_N}\right)$$

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

Student l_0: $\mathbf{o}_{l_0} = [0,1,1]$

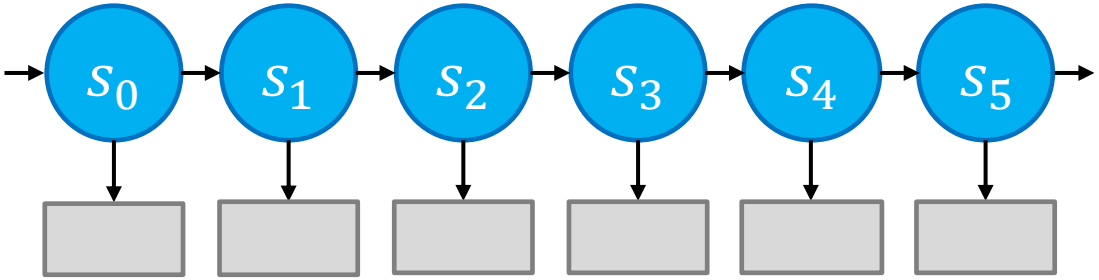
$\vdots$

Student l_{N-1}: $\mathbf{o}_{l_{N-1}} = [1,0,1,1,1,0,0,1,1,1]$

Student l_N: $\mathbf{o}_{l_N} = [0,1,0,1]$

$$\max_{\theta} \prod_{i=1}^{N} p(\boldsymbol{o}_{l_i})$$

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

Student l_0: $p(\mathbf{o}_{l_0}) = \sum_{\boldsymbol{s}} p(\mathbf{o}_{l_0}, \boldsymbol{s})$

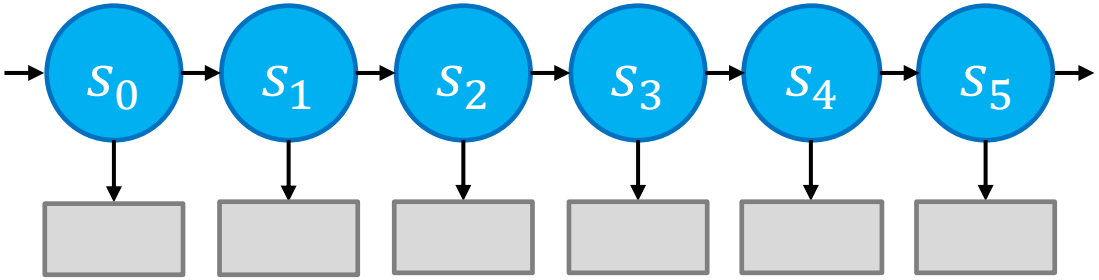
⋮

Student l_{N-1}: $p(\mathbf{o}_{l_0}) = \sum_{\boldsymbol{s}} p(\mathbf{o}_{l_0}, \boldsymbol{s})$

Student l_N: $p(\mathbf{o}_{l_0}) = \sum_{\boldsymbol{s}} p(\mathbf{o}_{l_0}, \boldsymbol{s})$

$$\max_{\theta} \prod_{i=1}^{N} \sum_{\boldsymbol{s}_{l_i}} p(\mathbf{o}_{l_i}, \boldsymbol{s}_{l_i})$$

Training a BKT model



$$\theta = \{p_0, p_L, p_F, p_S, p_G\}$$

$$\max_{\theta} \prod_{i=1}^{N} \sum_{\boldsymbol{s}_{l_i}} p(\mathbf{o}_{l_i}, \boldsymbol{s}_{l_i}) \Rightarrow \min_{\theta} -\sum_{i=1}^{N} \log\left(\sum_{\boldsymbol{s}_{l_i}} p(\mathbf{o}_{l_i}, \boldsymbol{s}_{l_i})\right)$$

- Brute-Force Grid Search
- Expectation Maximization
- Gradient Descent
- Nelder-Mead Optimization

Summary – Why tracing knowledge?

- Is the student learning?
  - Measure what the student *knows* at a specific time *t*
  - More specifically: knowledge of the student about relevant knowledge components (skills)

➡ Choose the next appropriate activity

➡ Know which activities support learning